

# **Week 1: overview / review**

**NRSC 7657 Workshop in Advanced Programming for  
Neuroscientists**

# Course outline

- Language of choice (python or MATLAB pref.)
- Didactic and practical
  - Language specifics
  - General concepts in computing (version control, debugging, unit testing, scaling)
- Independent project
  - Work with real data; preferably your own (whatever it is!) or choose a public data set (see next slide)
  - Develop an idea; **\*\*Schedule a meeting with Dan this week\*\***

<https://zcal.co/i/TnPIYgEZ>

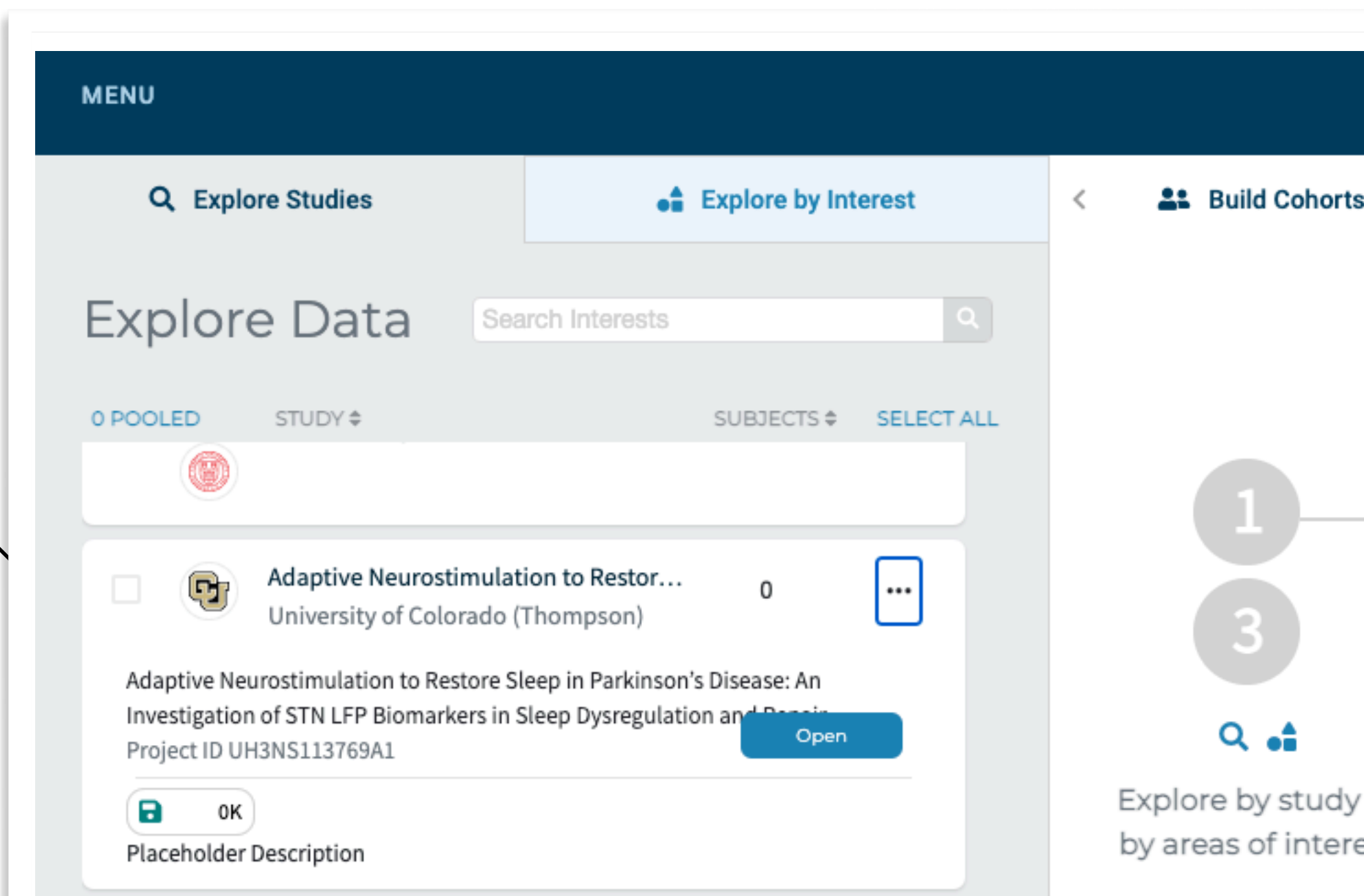
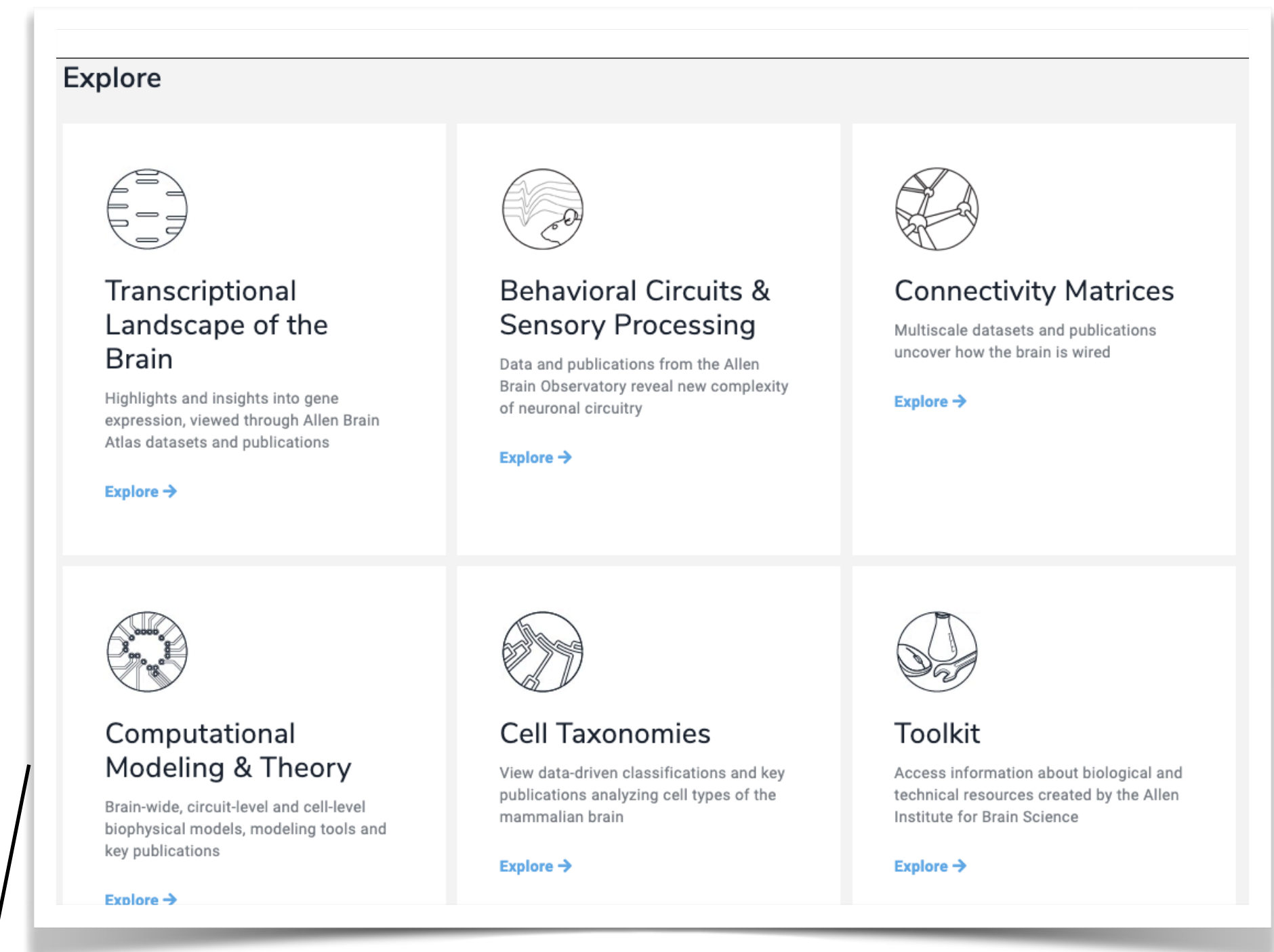
|           |                                               |                                                                                                                                                                                                                             |
|-----------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 9    | Week 1 – overview / review                    | Course overview: theory of computing, landscape of computing options.<br>Basic usage in python and MATLAB; basic data types; environments<br>Style guidelines ( <a href="#">ten simple rules</a> ); git and version control |
| June 16   | Week 2 – language fundamentals                | Functions; Objects and Classes; Workspaces<br>Typical data formats: working with tabular data, images, and time series.<br>NeurodataWithoutBorders format                                                                   |
| June 23   | Week 3 – workflow management and outputs      | Importing and exporting<br>Packaging                                                                                                                                                                                        |
| June 30   | Week 4 – usability                            | Plotting and visualization - from bar charts to 3D animation                                                                                                                                                                |
| July 7    | Week 5 – scaling                              | Troubleshooting and debugging; unit testing                                                                                                                                                                                 |
| July 14   | Week 6 – collaboration                        | Iteration and code profiling; parallel computing. Code quality-of-life topics                                                                                                                                               |
| July 21   | Week 7 – applications/flex topic              | Cloud-based tools: AWS, GCC, Colab, jupyterhub, deepnote.<br>Overview of some available SAAS tools, python focused.                                                                                                         |
|           |                                               | Group programming time                                                                                                                                                                                                      |
| July 28   | Week 8 – Open topics                          | Make-up session – open topics                                                                                                                                                                                               |
| August 4  | Week 9 – Final presentations and code review  | Final pres. and code review<br>Final pres. and code review                                                                                                                                                                  |
| August 11 | Week 10 – Final presentations and code review | Final pres. and code review<br>Final pres. and code review                                                                                                                                                                  |

# Project datasets

- Yours!
- Someone from your lab

## Public datasets (Suggestions, incomplete list)

- Allen Institute for Brain Science [brain-map.org](https://brain-map.org)
- BRAIN Initiative archives
  - DABI <https://dabi.loni.usc.edu/home>
  - DANDI, <https://dandiarchive.org/>





# What this course is **\*\*not\*\***

- Computational neuroscience
- Mathematics
- Computer science





# What this course is

## Each session:

- Some slides (0 - 60 minutes), a notebook on your own time
- Working through the notebook or interactive coding together. “Flipppped” classroom each week (after this week). So what we do depends on you.
- Independent coding. We’re going to do your science, with a computer. Maybe your computer, or maybe a cloud computer. Definitely your science.

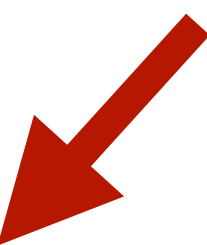
**content for each week available Tuesday AM for the next Monday  
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**Ask questions!**



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# Goals of this course

- Exposure to differing approaches to computing in neuroscience
- Develop confidence in independent coding skills
- Complete a project using data relevant to your thesis work



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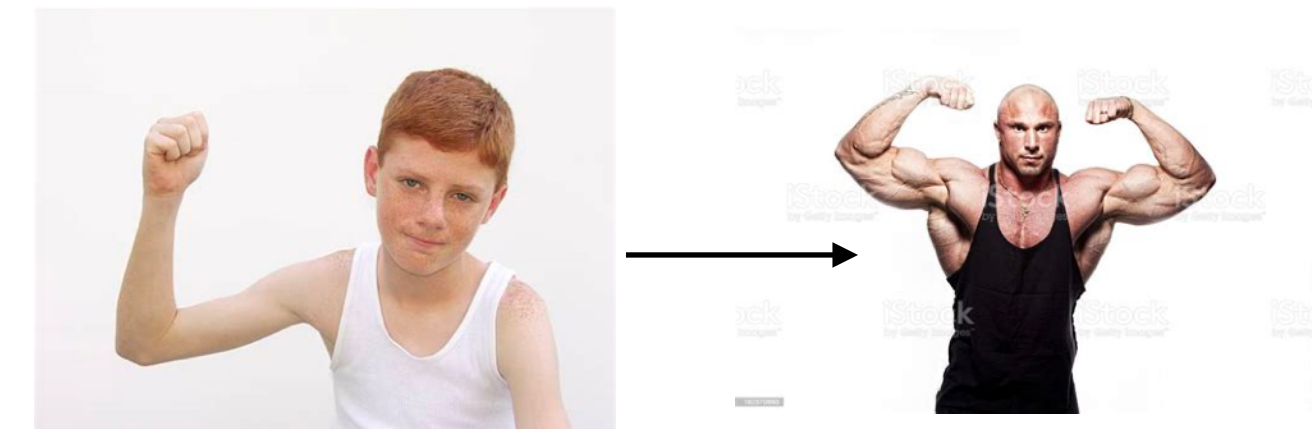


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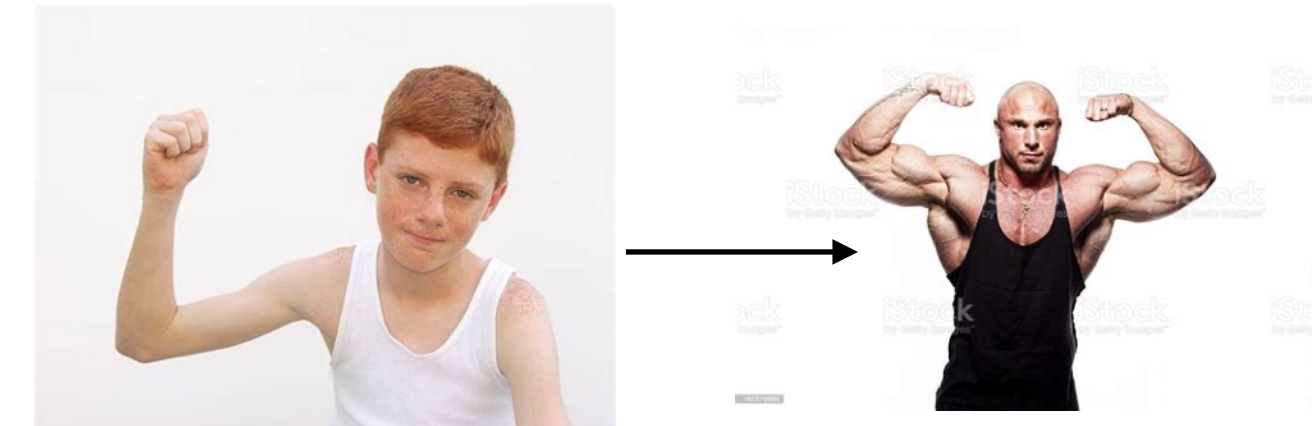
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- Develop confidence in independent coding skills



- Complete a project using data relevant to your thesis work





# Final practicalities

- GitHub: <https://github.com/CU-NRSC-7657/NRSC7657>

The screenshot shows the GitHub repository page for **CU-NRSC-7657 / NRSC7657**. The repository has 1 branch (main) and 0 tags. The commit history shows a recent update to README.md by danieljdenman 7 hours ago. The file list includes Week1, LICENSE, NRSC 7657 Workshop in Advanced ..., and README.md. The README.md file is open, showing the title **NRSC 7657, Workshop in Advanced Programming for Neuroscientists**, the affiliation **University of Colorado School of Medicine, Anschutz Medical Campus**, and the author **Daniel J Denman, 11 June 2021**. The right sidebar contains sections for About, Releases, Packages, and Languages.

Search or jump to... Pull requests Issues Marketplace Explore

CU-NRSC-7657 / NRSC7657 Unwatch 1 Star 0 Fork 0

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main 1 branch 0 tags Go to file Add file Code

danieljdenman Update README.md 831f1b2 7 hours ago 5 commits

|                                    |                  |             |
|------------------------------------|------------------|-------------|
| Week1                              | add week 1       | 4 days ago  |
| LICENSE                            | Initial commit   | 4 days ago  |
| NRSC 7657 Workshop in Advanced ... | add week 1       | 4 days ago  |
| README.md                          | Update README.md | 7 hours ago |

README.md

## NRSC 7657, Workshop in Advanced Programming for Neuroscientists

University of Colorado School of Medicine, Anschutz Medical Campus

Daniel J Denman, 11 June 2021

### About

No description, website, or topics provided.

Readme

GPL-3.0 License

### Releases

No releases published  
[Create a new release](#)

### Packages

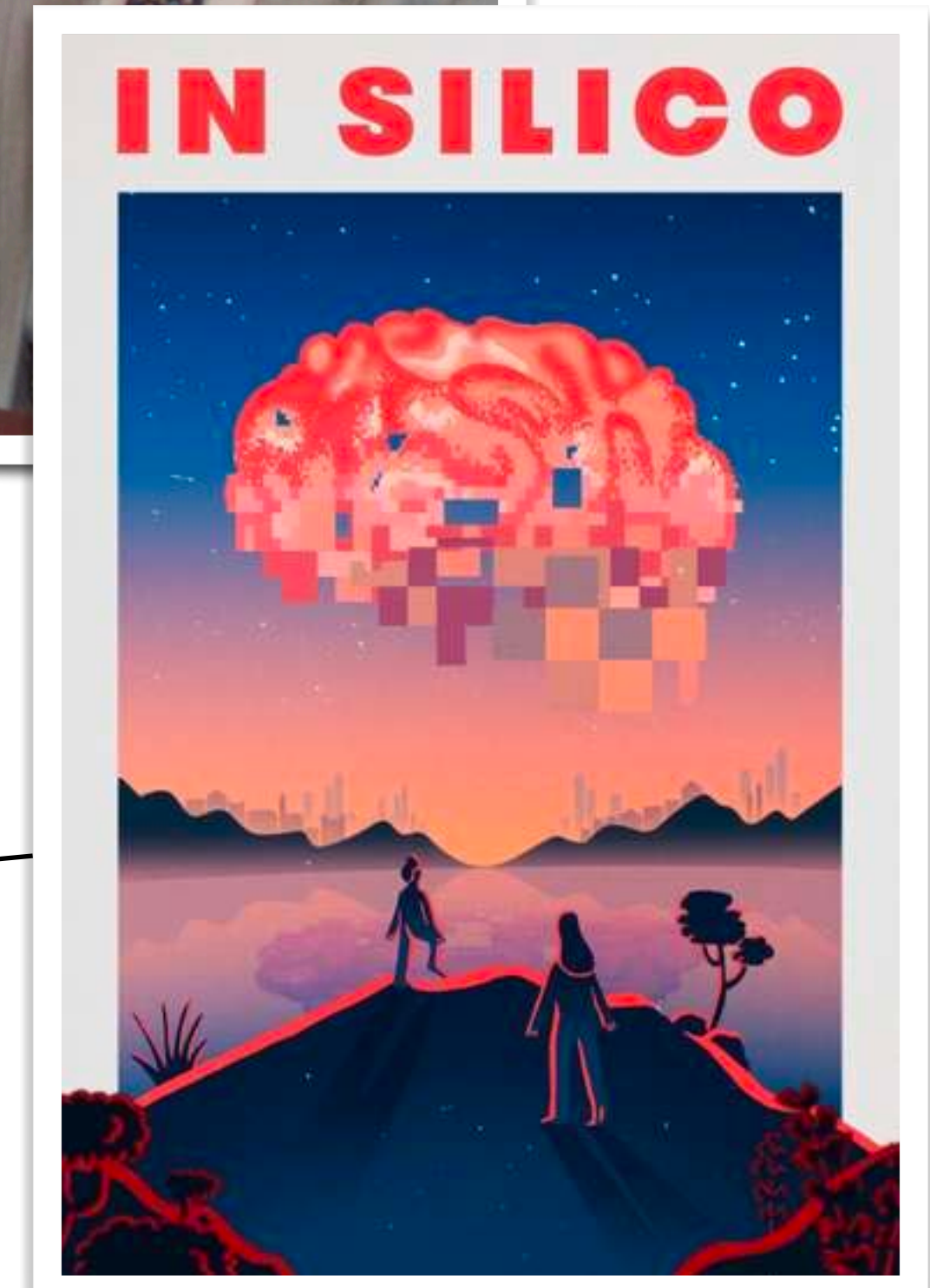
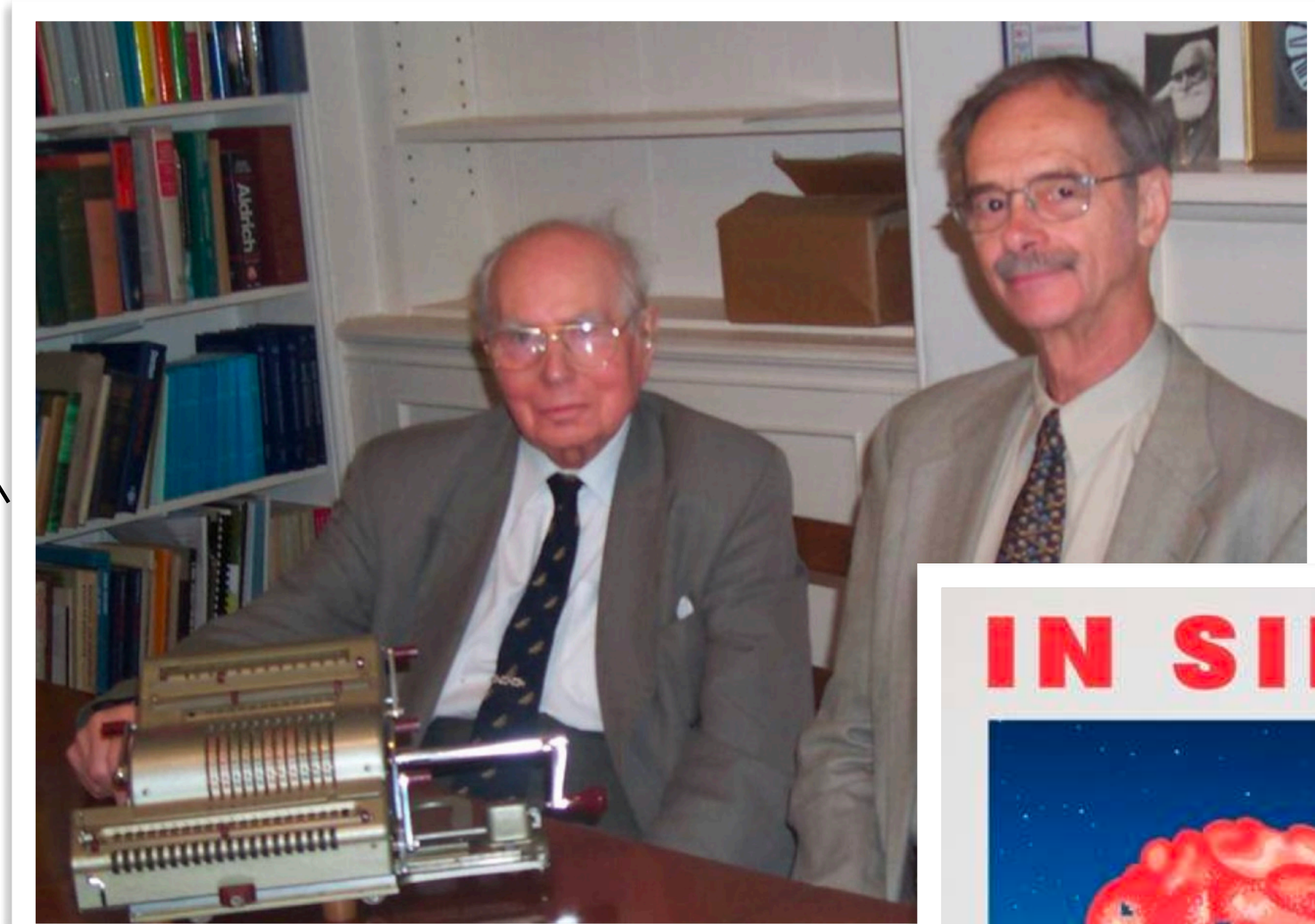
No packages published  
[Publish your first package](#)

### Languages



# Computing in neuroscience

- Hodkin-Huxley's mechanical computer
- Computer based analysis / replacing chart measurements
- NEURON simulations; single neuron modeling
- ...
- “data science”, open data
- Blue brain project, *In silico*





# Computing in neuroscience

## An example: code written by neuroscientists for one experiment

### planning

Probe trajectory: MATLAB  
Model prediction: python -  
jupyter notebook

### experimental control

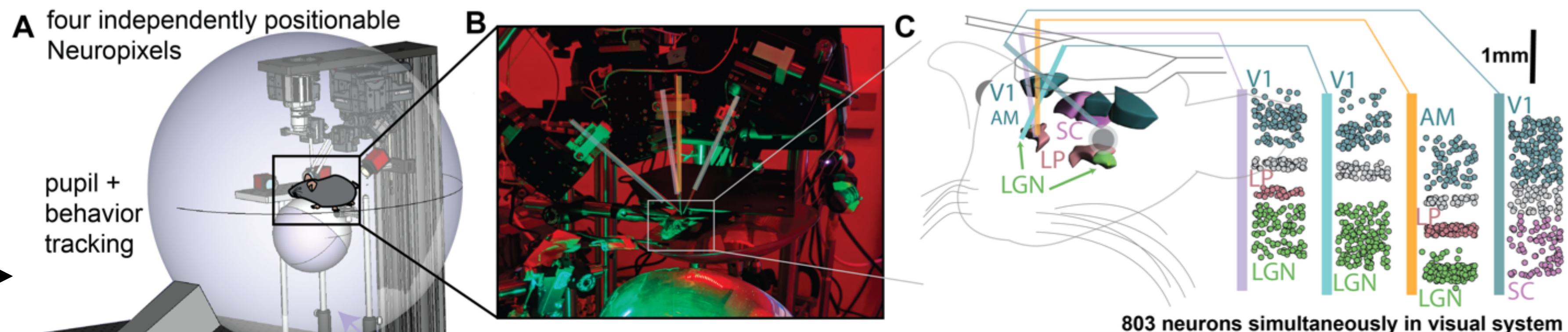
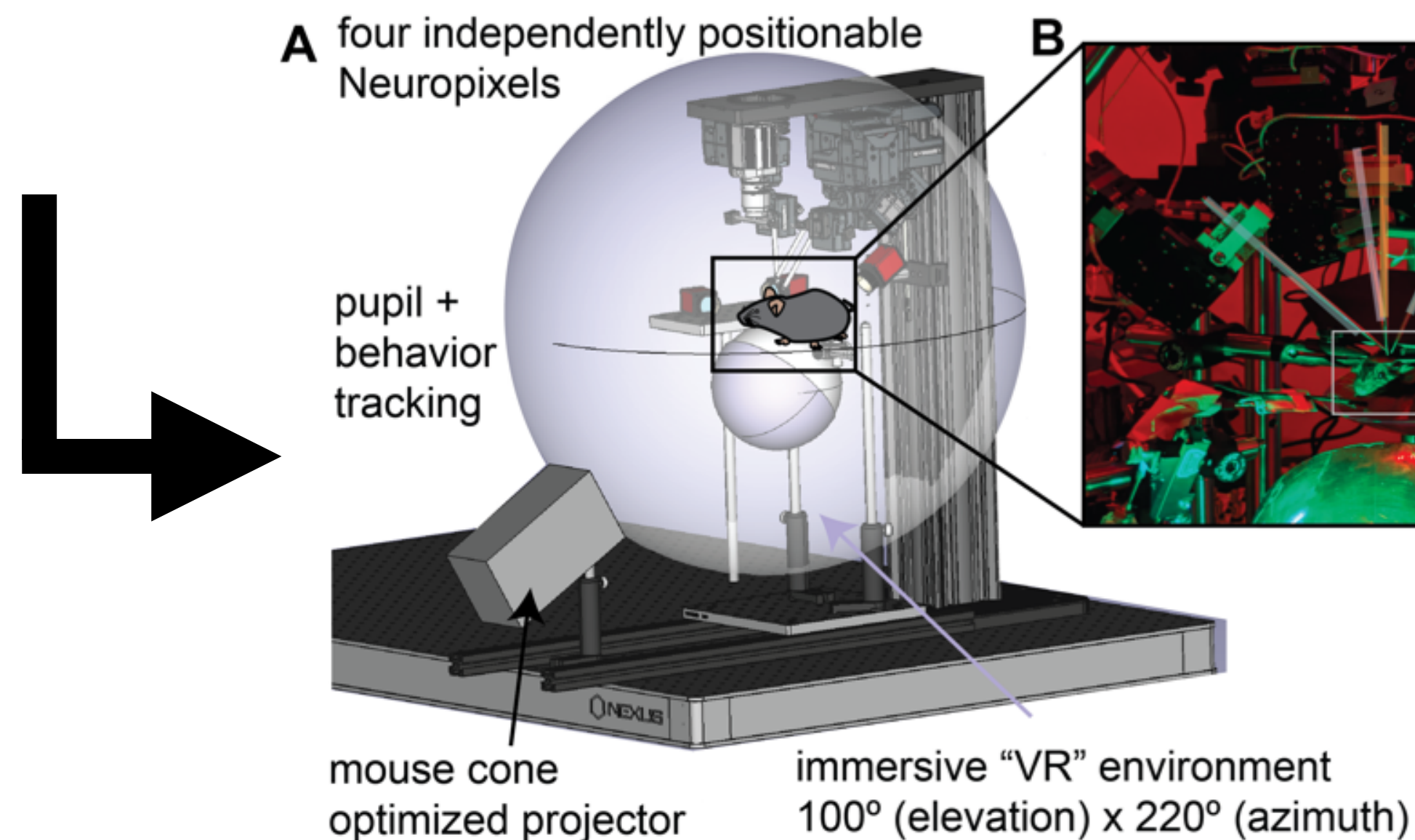
Acquisition hardware: FPGA gate array programming  
Acquisition software: C++; python plugin; [Julia]  
Visual stimuli: python script; embedded python in React  
Video monitoring: python script

### “pre-processing”

Spike sorting: MATLAB  
Unit quality: python - jupyter notebook  
Depth: python - jupyter notebook

### analysis

Histology registration to 3D  
brain: python  
Stimulus responses: python  
Population statistics: MATLAB  
and python



**Figure 1. Platform for studying distributed visual coding.** A., Immersive visual environment for mouse psychophysics. B., Simultaneous Neuropixels recordings in the mouse visual system, spanning V1, cortical area AL, LGN thalamus, LP thalamus, and superior colliculus. C., 803 single neurons isolated from the above brain areas; anatomy renderings from the Allen Reference Atlas (brain-map.org).



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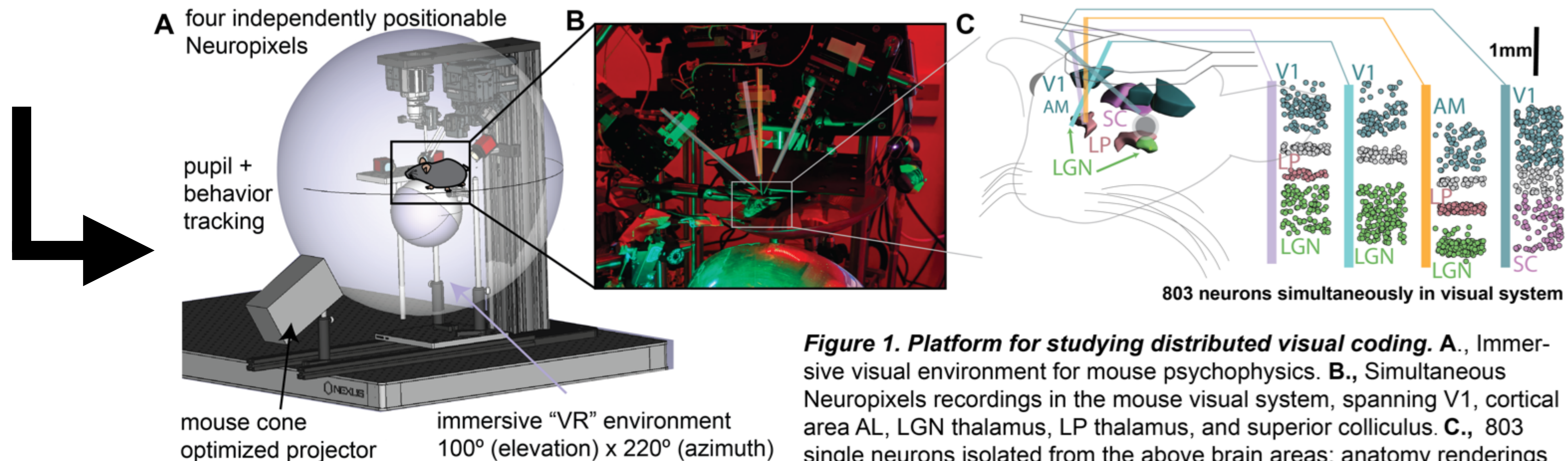
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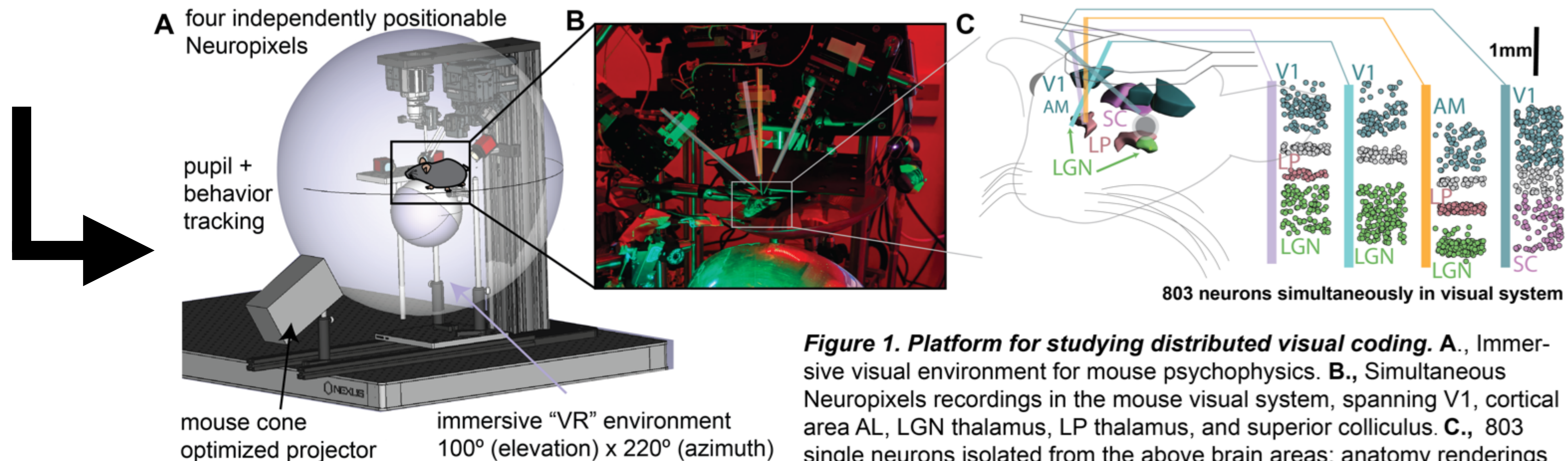
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# overview

Some plusses

Some minuses

# overview

## Some plusses

- Free
- Readable syntax
- Cross platform
- Huge community
- Used across science *and* outside of science

## Some minuses

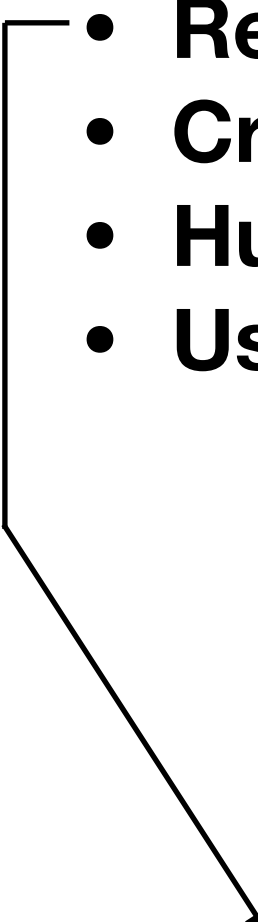


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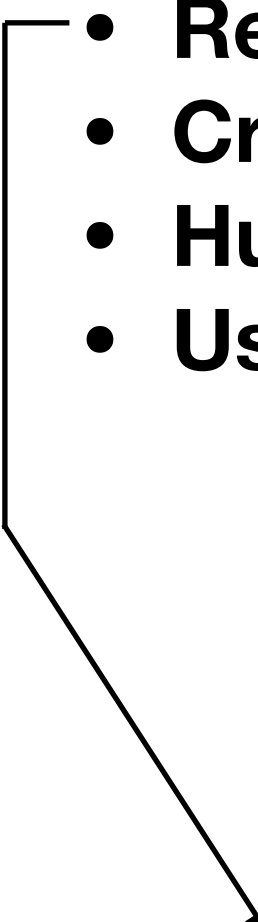
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Week 1

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# overview: packages

Find, install and publish Python packages  
with the Python Package Index



Or [browse projects](#)

381,490 projects

3,536,531 releases

6,207,207 files

600,160 users

## Packages

In addition to the standard library, the true power of python is the extensive world of packages available. These are sets of tools you can use with Python to do just about anything!

Some are general tools, the hammers or screwdrivers of using python for science:  
**numpy, matplotlib, pandas, seaborn**

Others are specialized: **scikit-learn, PIL, scanpy, Suite2P, DeepLabCut, PyTom**





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Week 3

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# overview: levels





## System

# overview: levels

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danieljdenman — python — 80x24
Last login: Wed Dec 18 16:26:32 on ttys002

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
(base) djd-mbpro:~ danieljdenman$ python
Python 3.7.4 (default, Aug 13 2019, 15:17:50)
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## Package Managers Environments





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## Containerized







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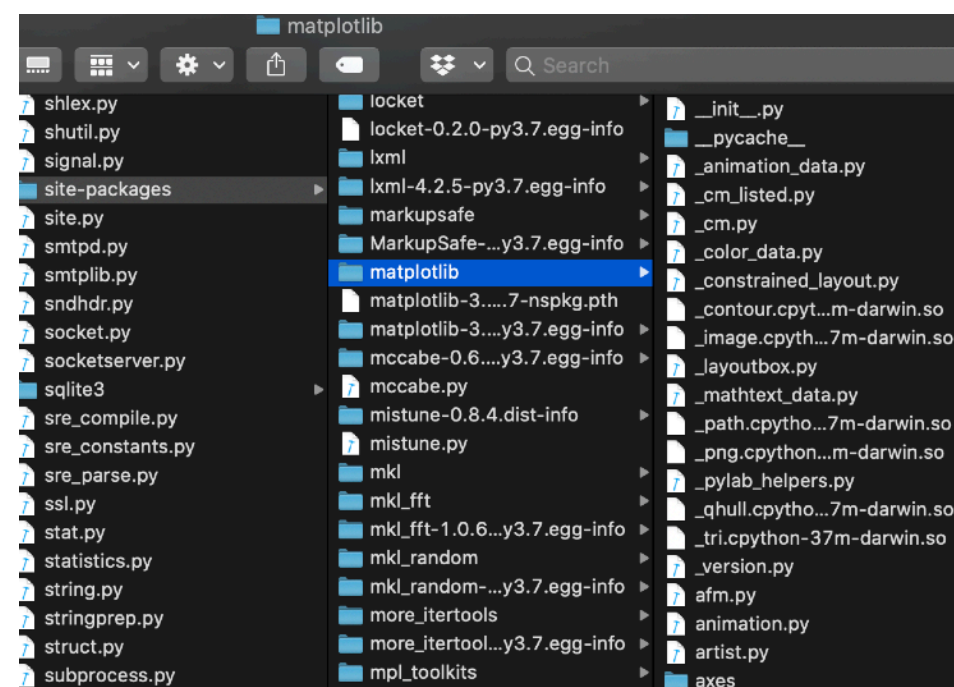
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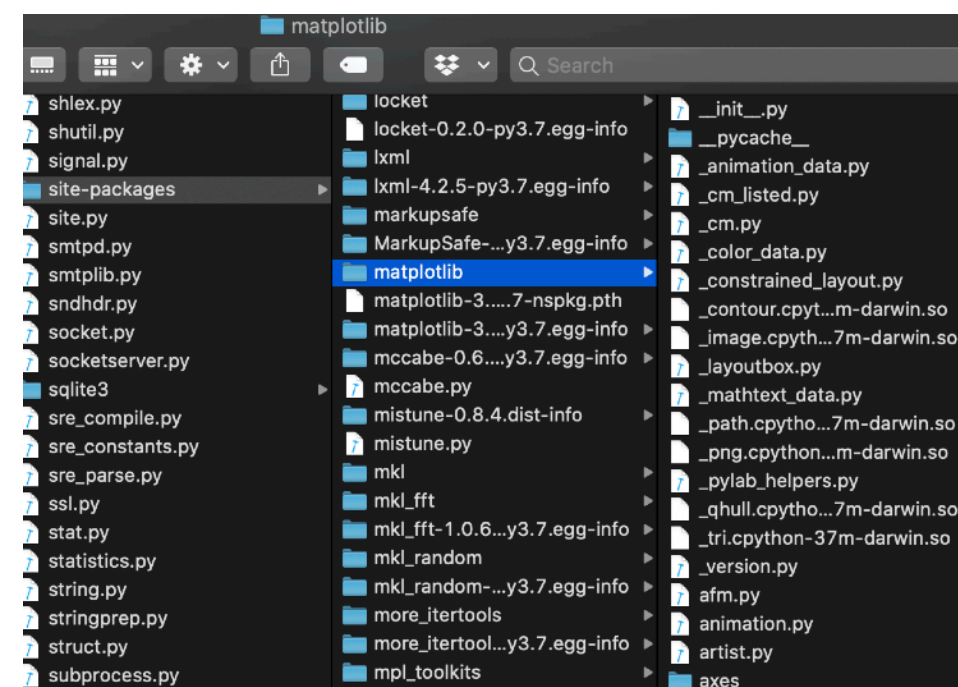
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4  # from direct.showbase.DirectObject import DirectObject
5  from direct.interval.MetaInterval import Sequence
6  from direct.interval.LerpInterval import LerpFunc
7  from direct.interval.FunctionInterval import Func
8  from panda3d.core import Mat4, WindowProperties, CardMaker, NodePath, TextureStage, MovieTexture, MovieVideo
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10 import sys,glob,time,datetime,os
11 from math import pi, sin, cos
12 from numpy.random import randint, exponential
13 from numpy import arange,concatenate
14 import numpy as np
15 from pyglet.window import key
16
17 try:
18     from toolbox.toolbox.IO.nidaq import DigitalInput,DigitalOutput, AnalogInput, AnalogOutput
19     have_nidaq=True
20 except:# Exception, e:
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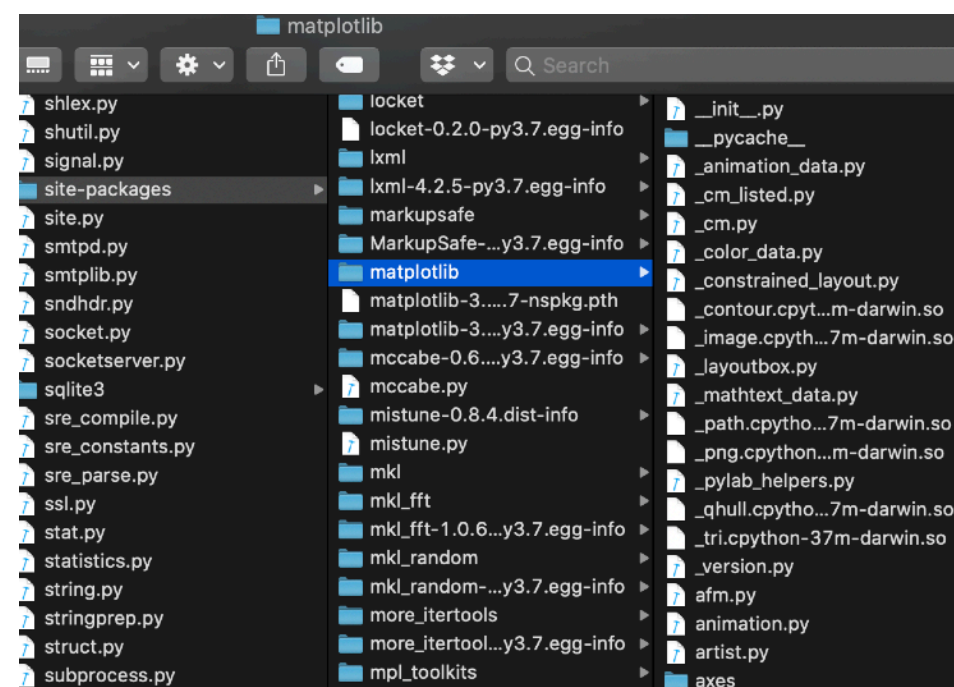
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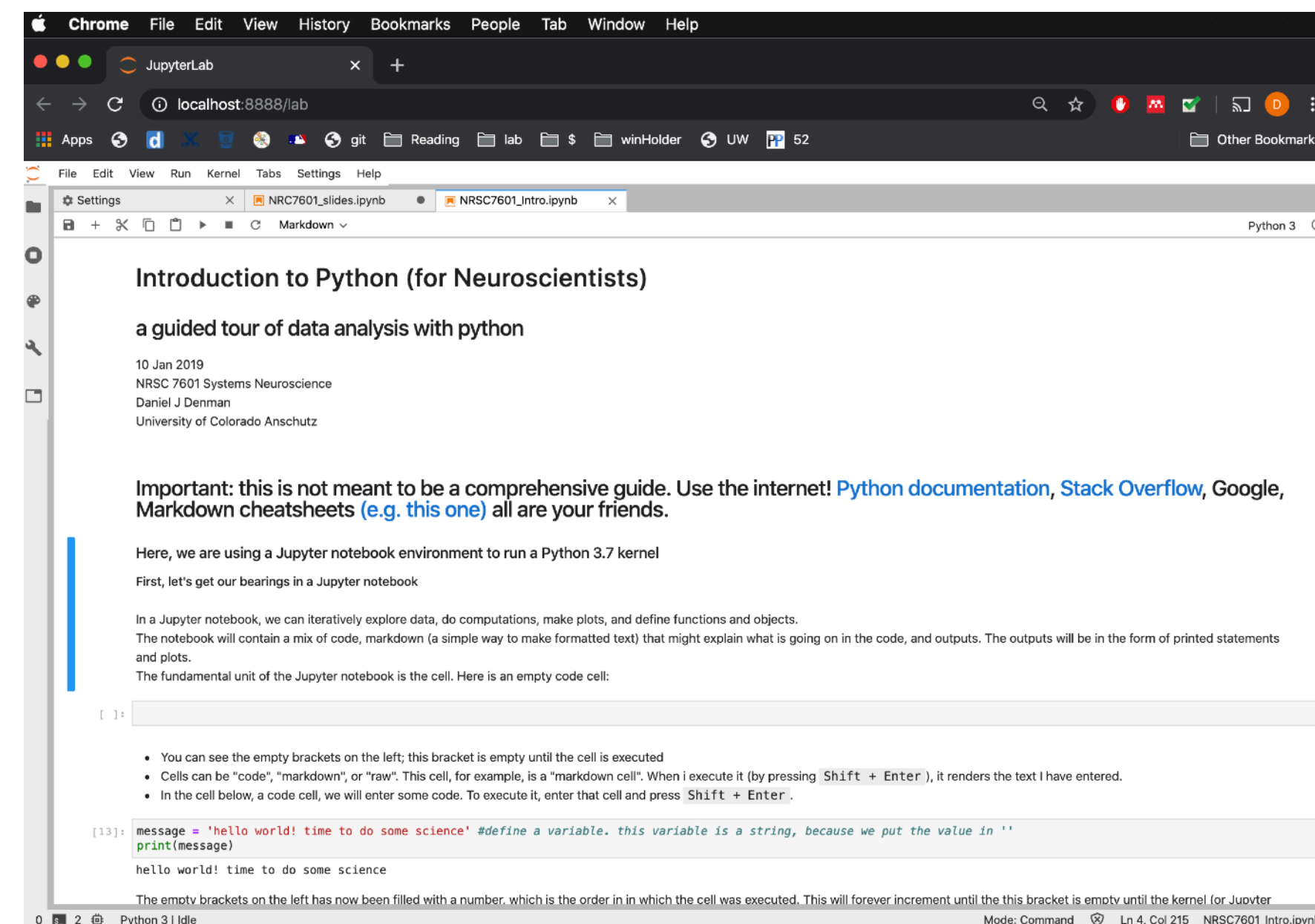
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## Containerized



## Notebooks (IPython, Jupyter, Jupyter Lab)







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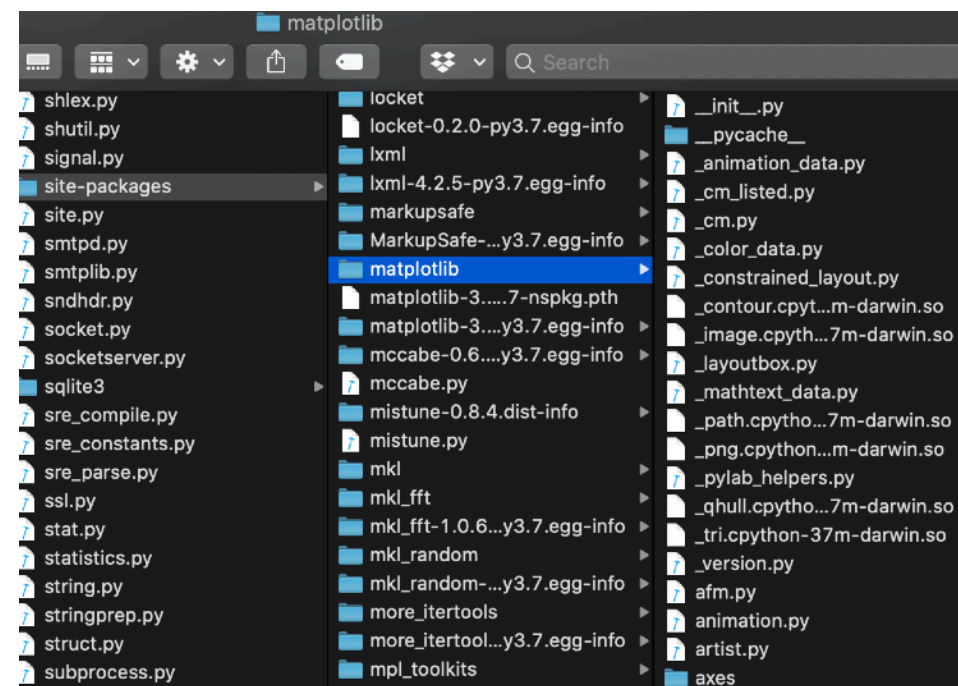
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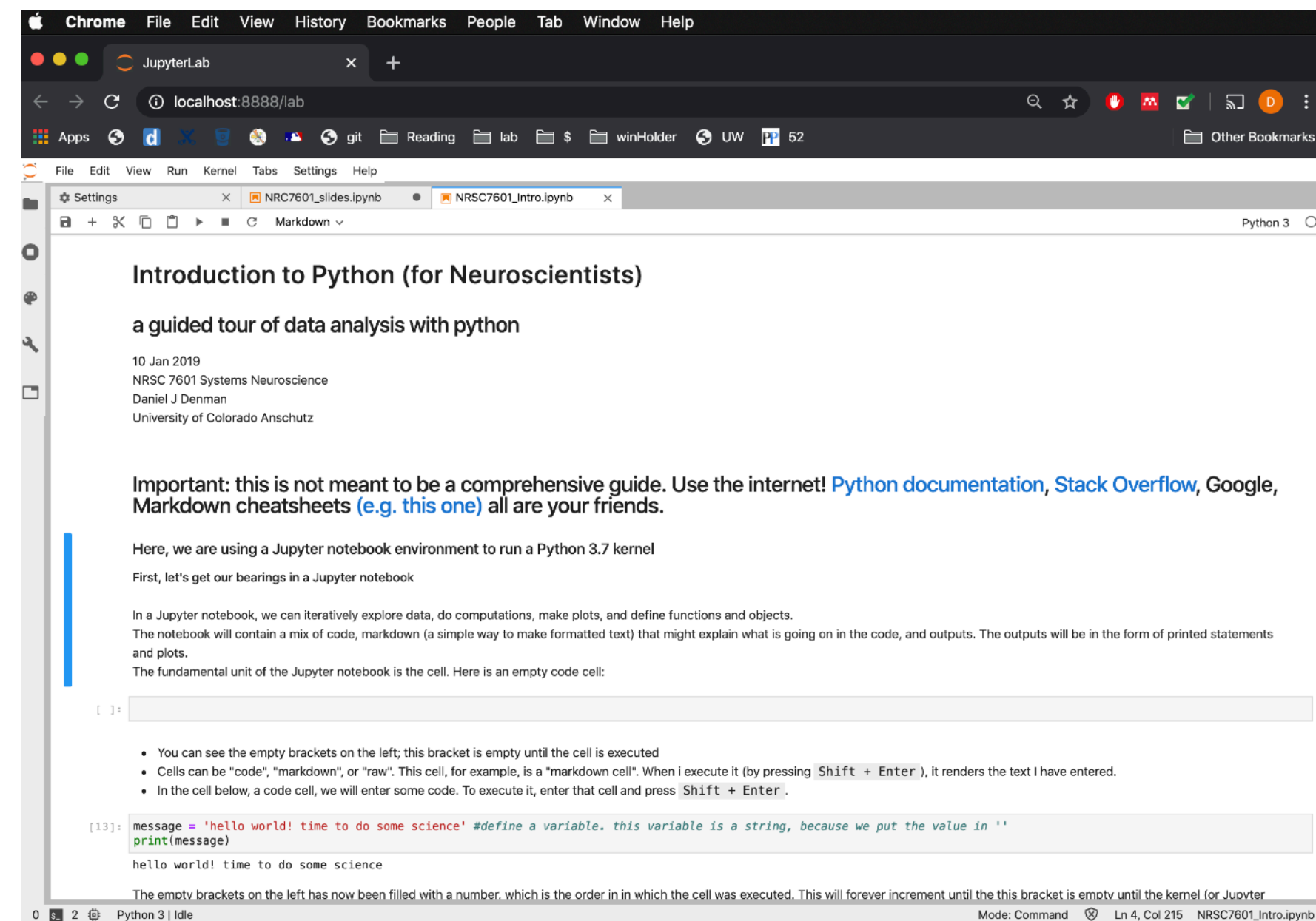
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## Packages



Week 1

## Notebooks (IPython, Jupyter, Jupyter Lab)







## System

```
danieljdenman — python — 80x24
Last login: Wed Dec 18 16:26:32 on ttys002

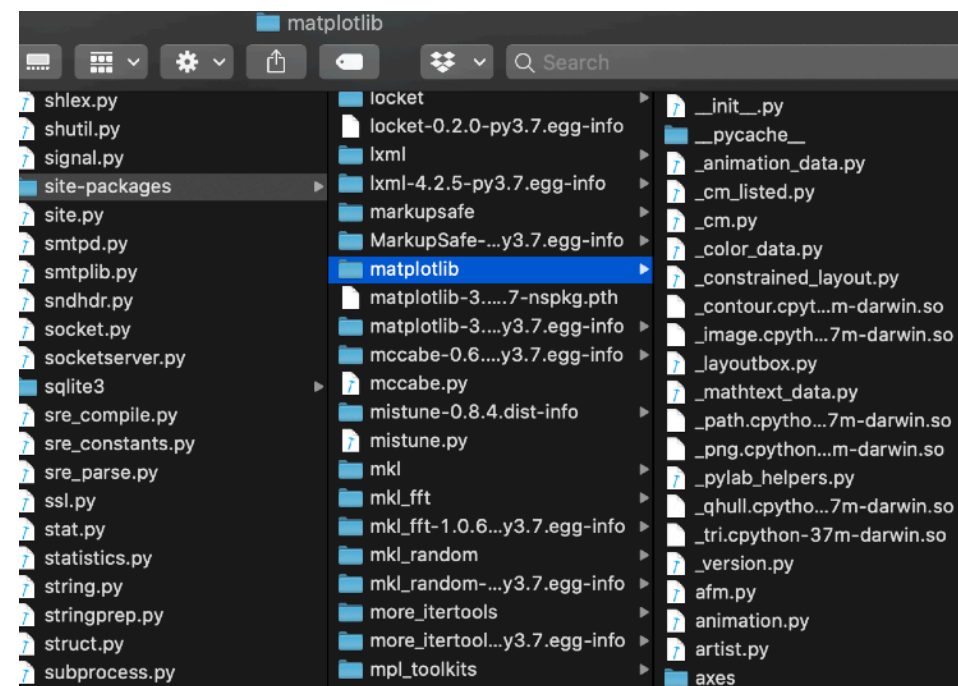
The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
(base) djd-mbpro:~ danieljdenman$ python
Python 3.7.4 (default, Aug 13 2019, 15:17:50)
[Clang 4.0.1 (tags/RELEASE_401/final)] :: Anaconda, Inc. on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Native in Mac OS X, Linux; in Windows store (free)

## Scripts

```
Users > danieljdenman > github > mouse_tunnel > mouse_tunnel_auto_CUtest.py
1 from direct.showbase.ShowBase import ShowBase
2 from direct.task import Task
3 # from direct.gui.OnscreenText import OnscreenText
4 # from direct.showbase.DirectObject import DirectObject
5 from direct.interval.MetaInterval import Sequence
6 from direct.interval.LerpInterval import LerpFunc
7 from direct.interval.FunctionInterval import Func
8 from panda3d.core import Mat4, WindowProperties, CardMaker, NodePath, TextureStage, MovieTexture, MovieVideo
9
10 import sys,glob,time,datetime,os
11 from math import pi, sin, cos
12 from numpy.random import randint, exponential
13 from numpy import arange,concatenate
14 import numpy as np
15 from pygamelet.window import key
16
17 try:
18     from toolbox.toolbox.IO.nidaq import DigitalInput,DigitalOutput, AnalogInput, AnalogOutput
19     have_nidaq=True
20 except:# Exception, e:
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## Packages



# overview: levels

## Package Managers Environments

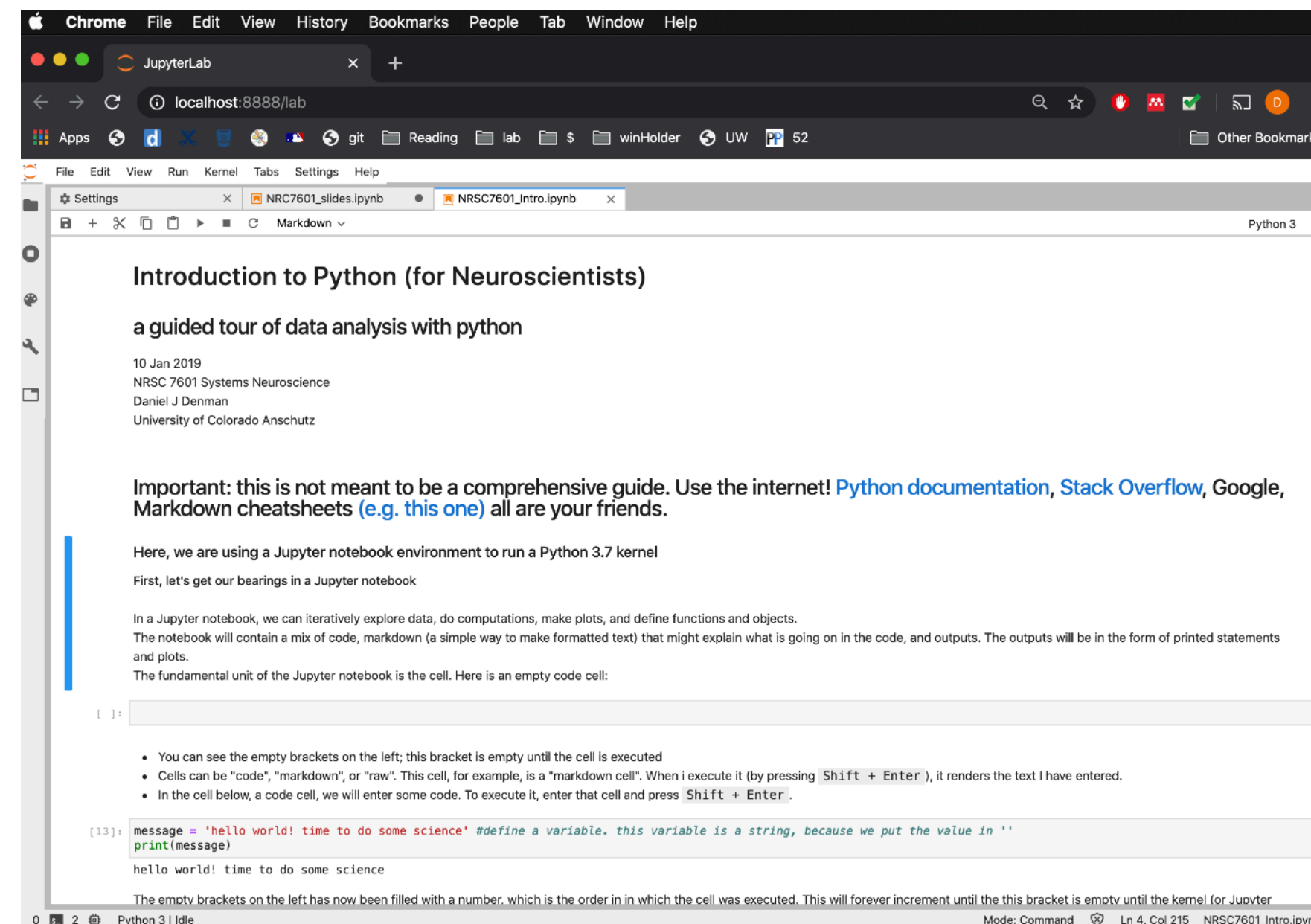


## Containerized

Week 6

Week 1

## Notebooks (IPython, Jupyter, Jupyter Lab)







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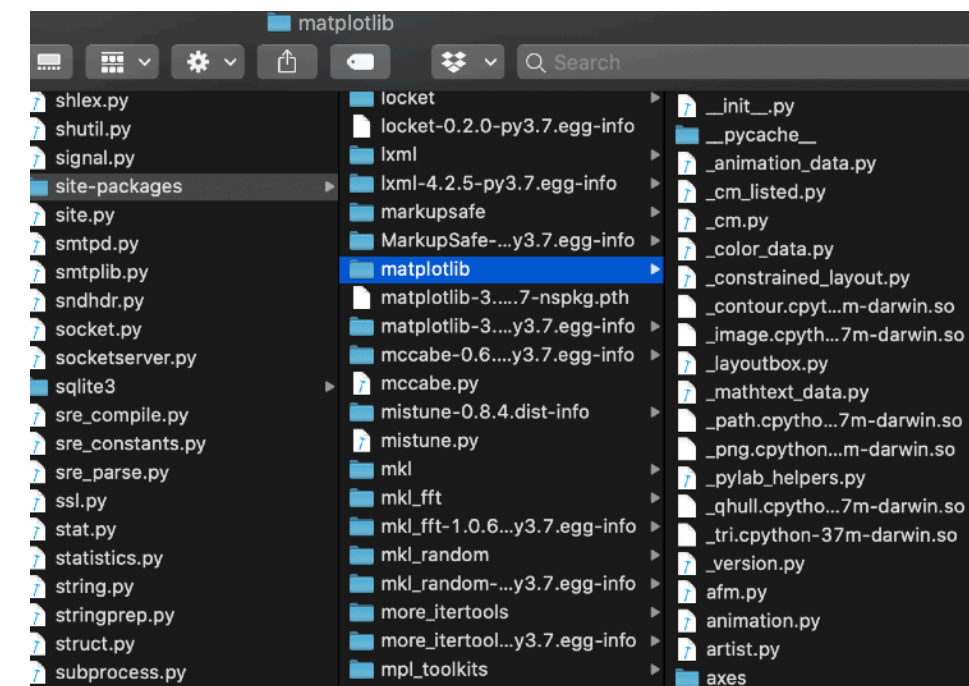
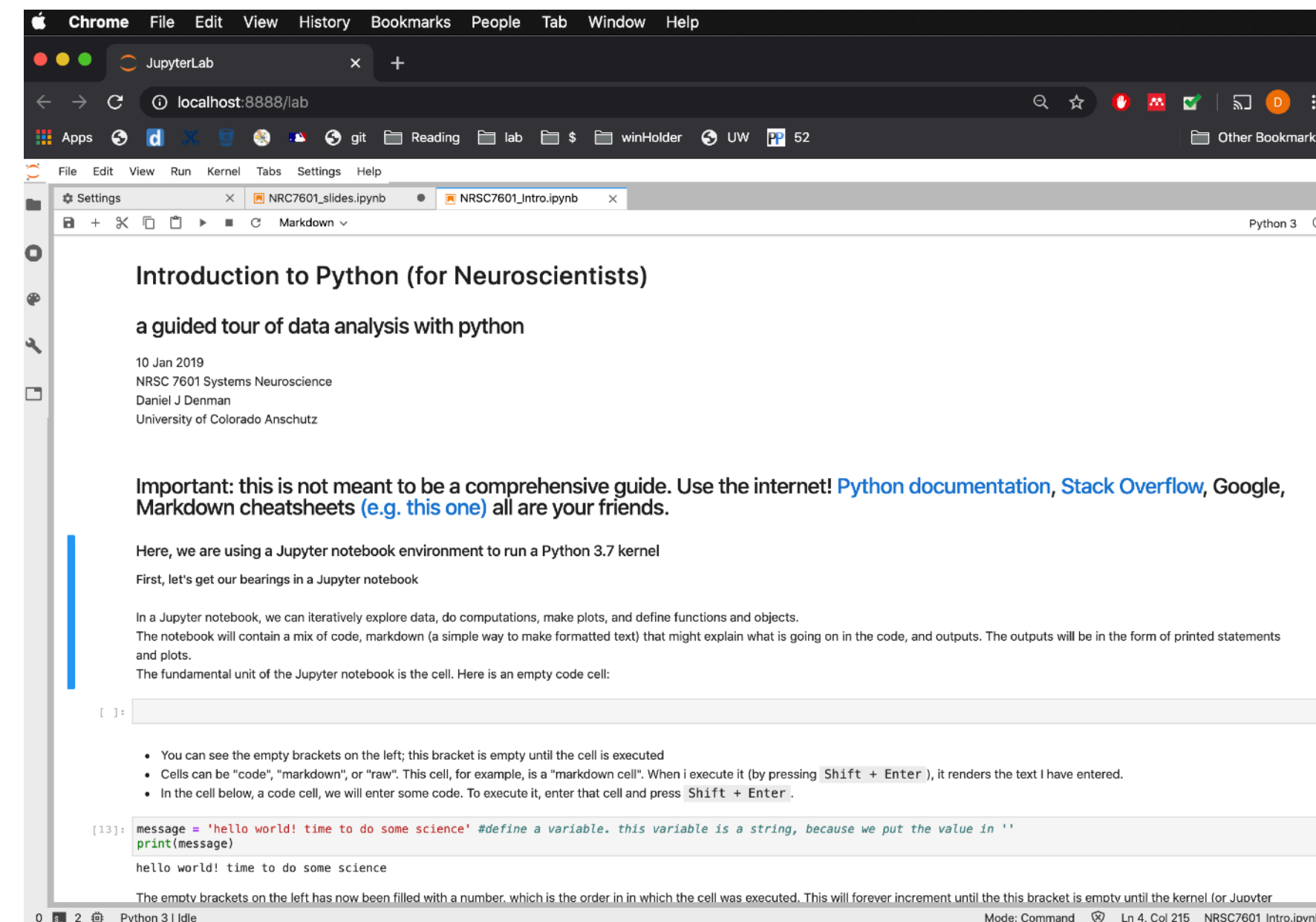
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## Package Managers Environments



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## Packages

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Week 6

Week 1

Week 3



**why is it good for doing neuroscience?**

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- Do analyses that would be a whole packages! PhD to implement yourself (i.e., ML)
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in this realm, cloud resources (data, compute) also open science to a wider group that aren't collecting their own data and running their own super computers

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**Week 2**



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**Week 2**

**Week 5**



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**Week 2**

**Week 5**

**Week 6**



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**Weeks 7-9**

**Week 2**

**Week 5**

**Week 6**



# MATLAB

- At this time, some understanding of both Python and MATLAB is extremely useful. We use both in the Denman Lab, but my primary expertise is in Python
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## **Week 1**

- **Anaconda environments and where python is on your computer**
- **VS Code, starting a jupyter notebook, running a script**
- **Basics and syntax review**



# Style and philosophy

- **Don't:** overoptimize; get it done first
- **Do:** document as you go
- Pay attention to variable names
- There is a slippery slope from the minimal functional code to unusable/unshareable code
- Read this PLoS paper this week!

PLOS COMPUTATIONAL BIOLOGY

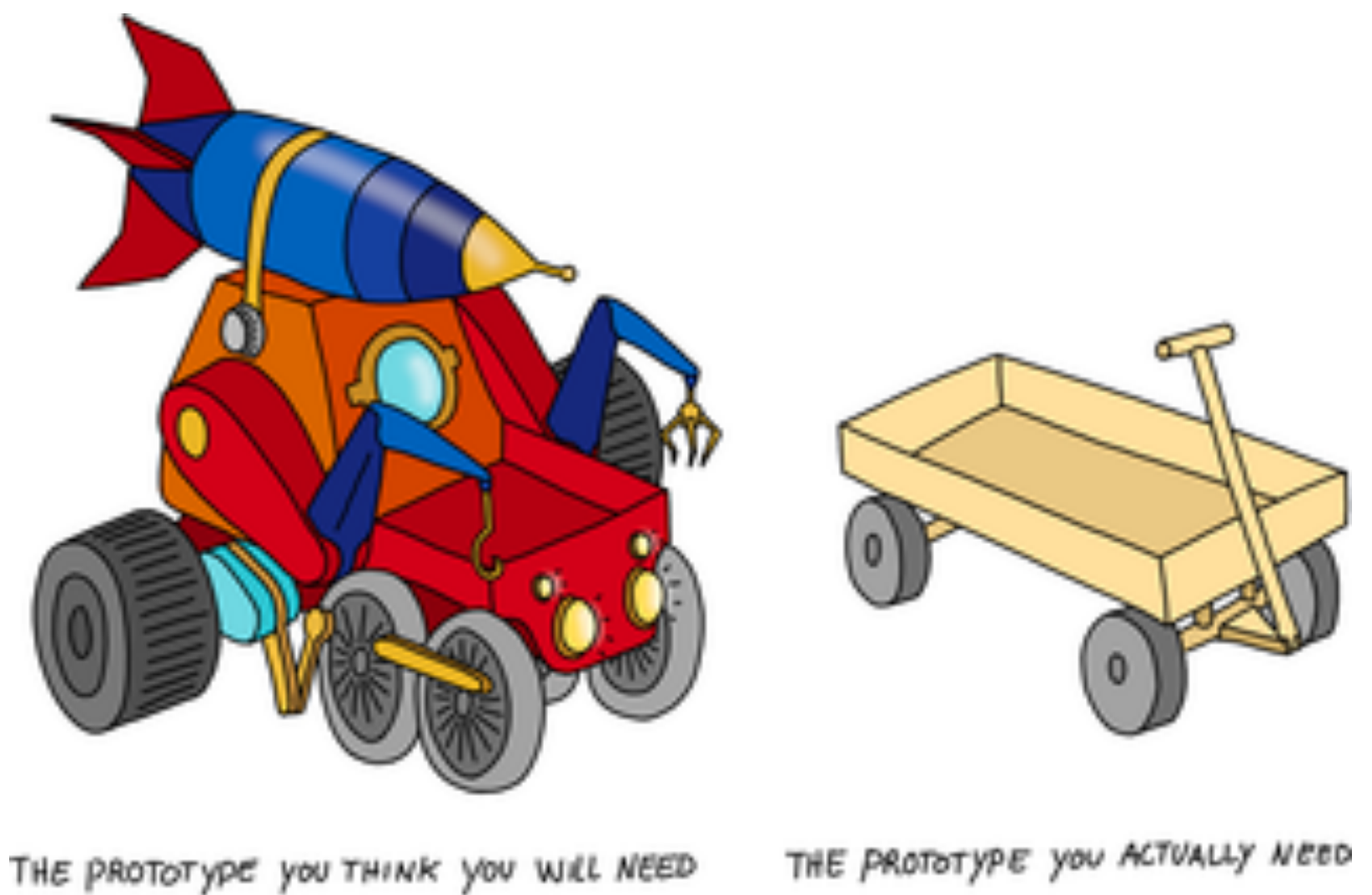
 OPEN ACCESS

EDITORIAL

Ten simple rules for quick and dirty scientific programming

Gabriel Balaban, Ivar Grytten, Knut Dagestad Rand, Lonneke Scheffer, Geir Kjetil Sandve 

Published: March 11, 2021 • <https://doi.org/10.1371/journal.pcbi.1008549>



<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1008549>

# git

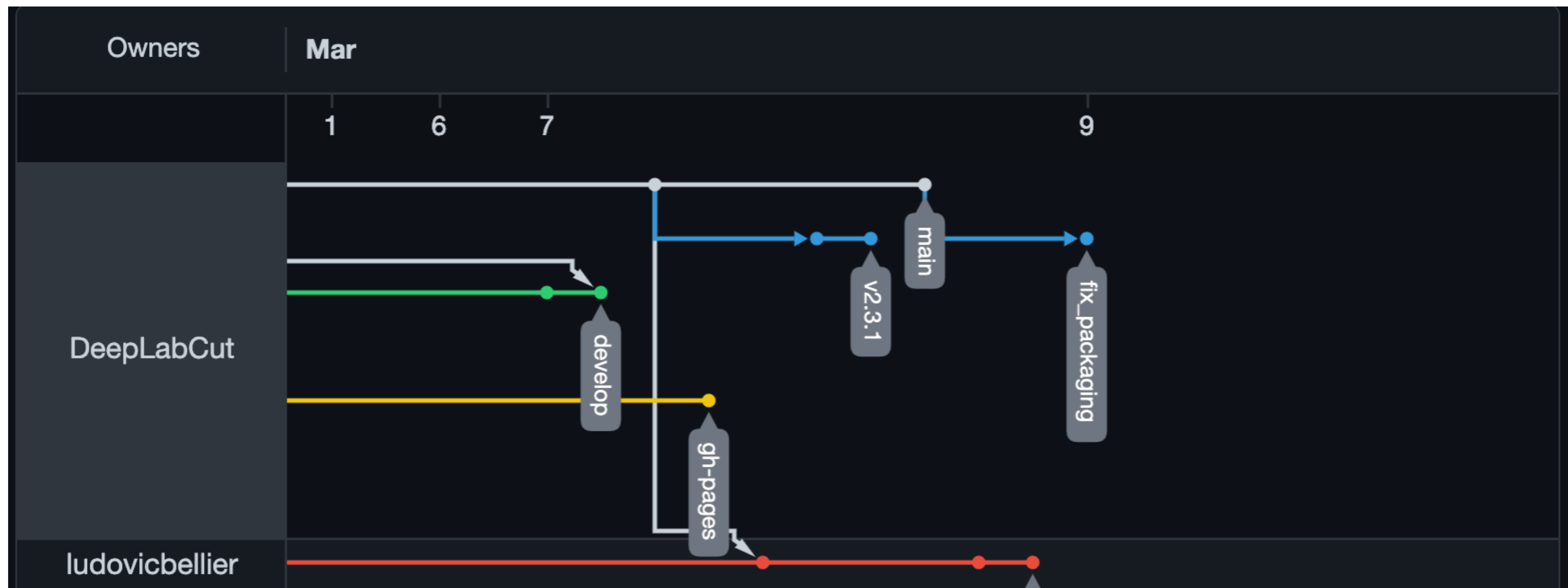
...and GitHub, are **version control**

- This is important. And not intuitive. It will likely make you frustrated and/or confused at some point.
- Version control is not optional; if you don't use git for version control, you are going to use something else (e.g.,: analysis\_script\_v1.py, analysis\_script\_v2.py, analysis\_script\_v2\_20210622.py, analysis\_script\_v3\_07142021.py, analysis\_script\_final.py, analysis\_script\_final2.py, ..., analysis\_script\_final2\_for.py)
- Making git a part of your workflow can simplify and provide redundancy and flexibility; more advanced features also makes sharing simpler. Evaluation.
- git has to be installed, which we can use Anaconda to do (even if you are using MATLAB only for your project)
- Command-line is great, but if you are new to git we recommend using VS Code or GitHub Desktop
- **We're going to go over some git interactively to get course materials today.**



# git

## Conceptually



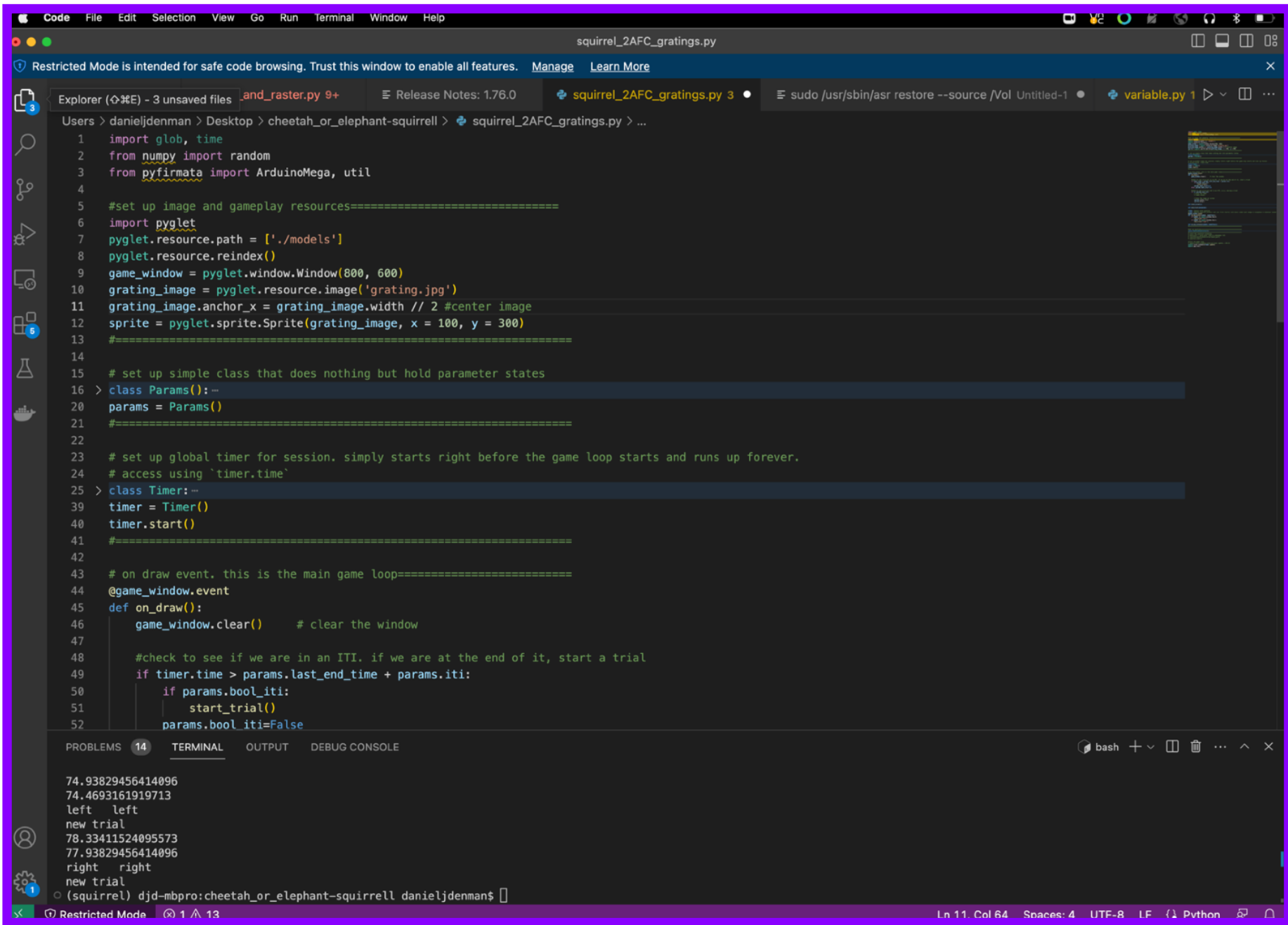
# GitHub

## Conceptually

Remote



Local





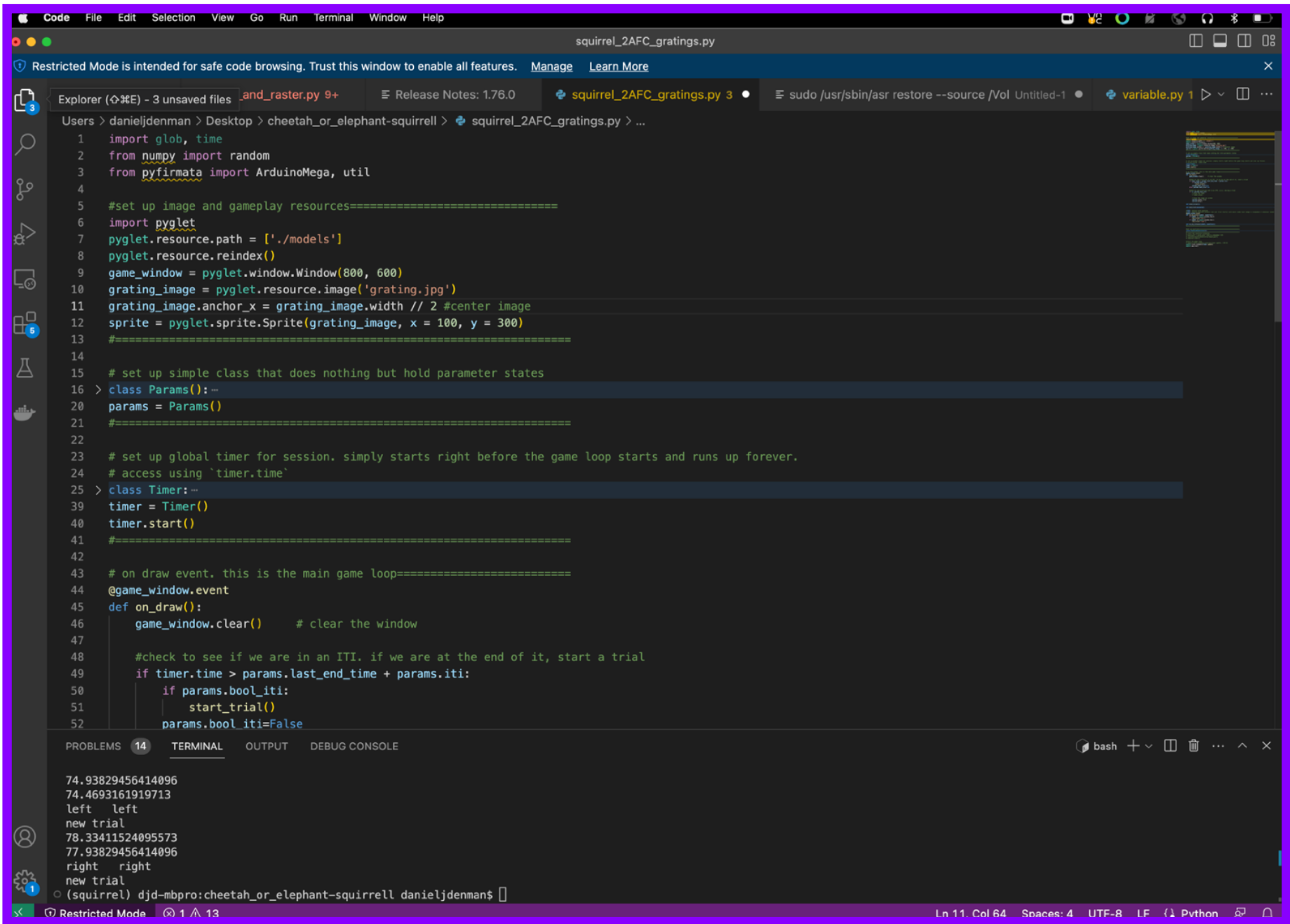
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Remote

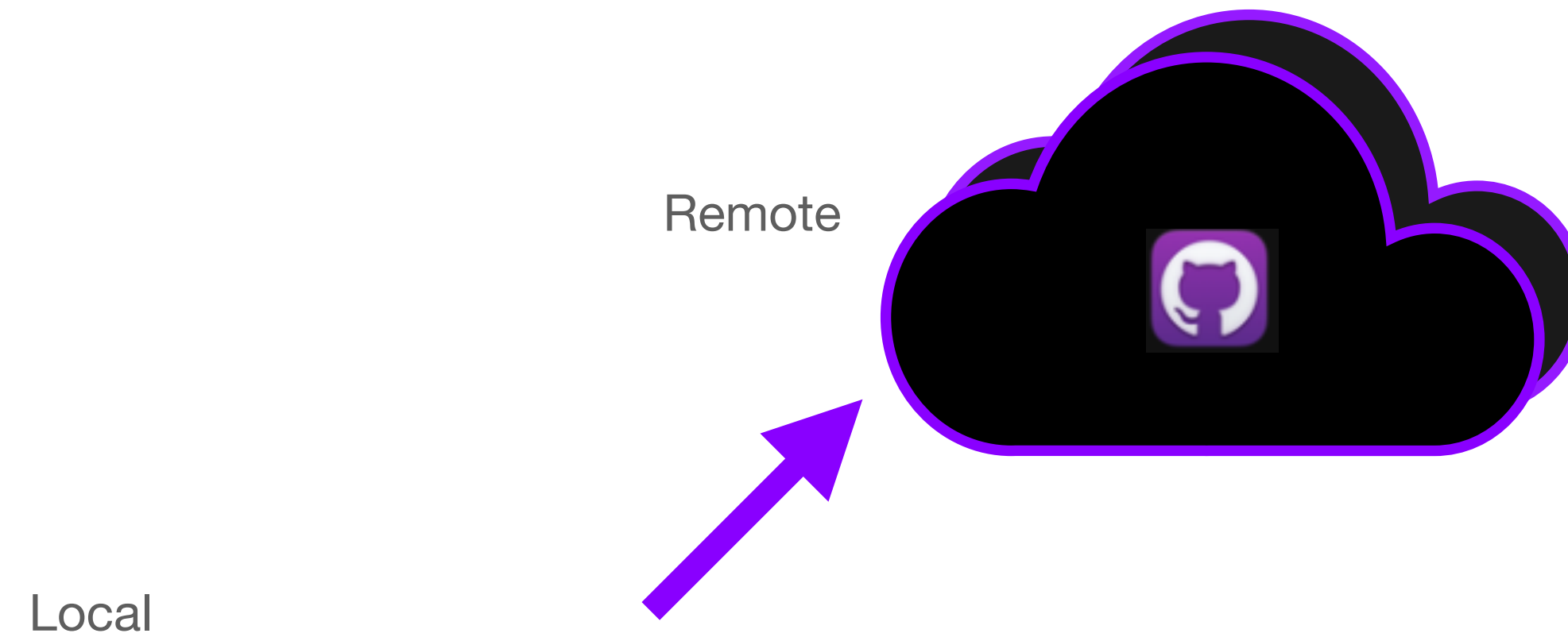


Local



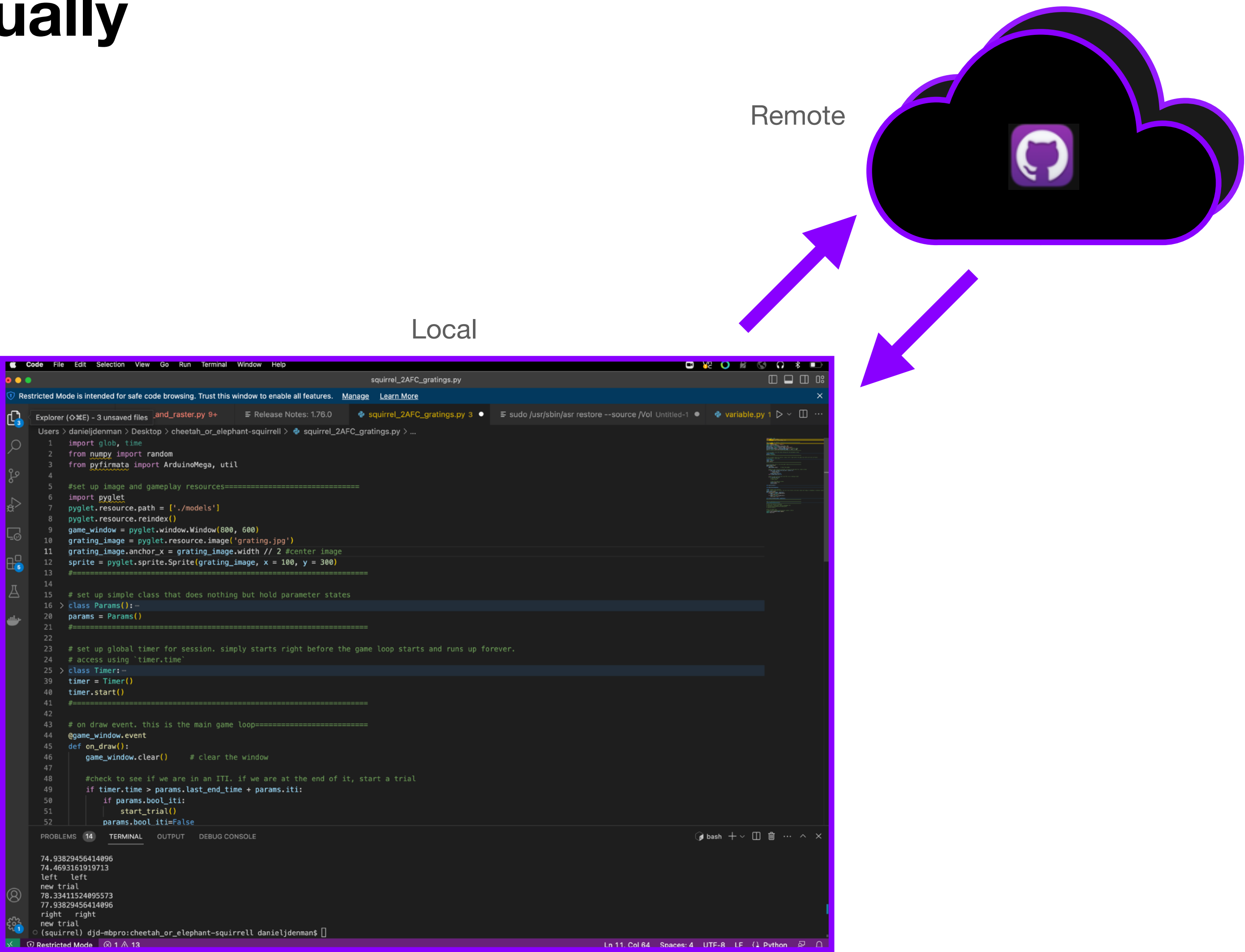
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# GitHub

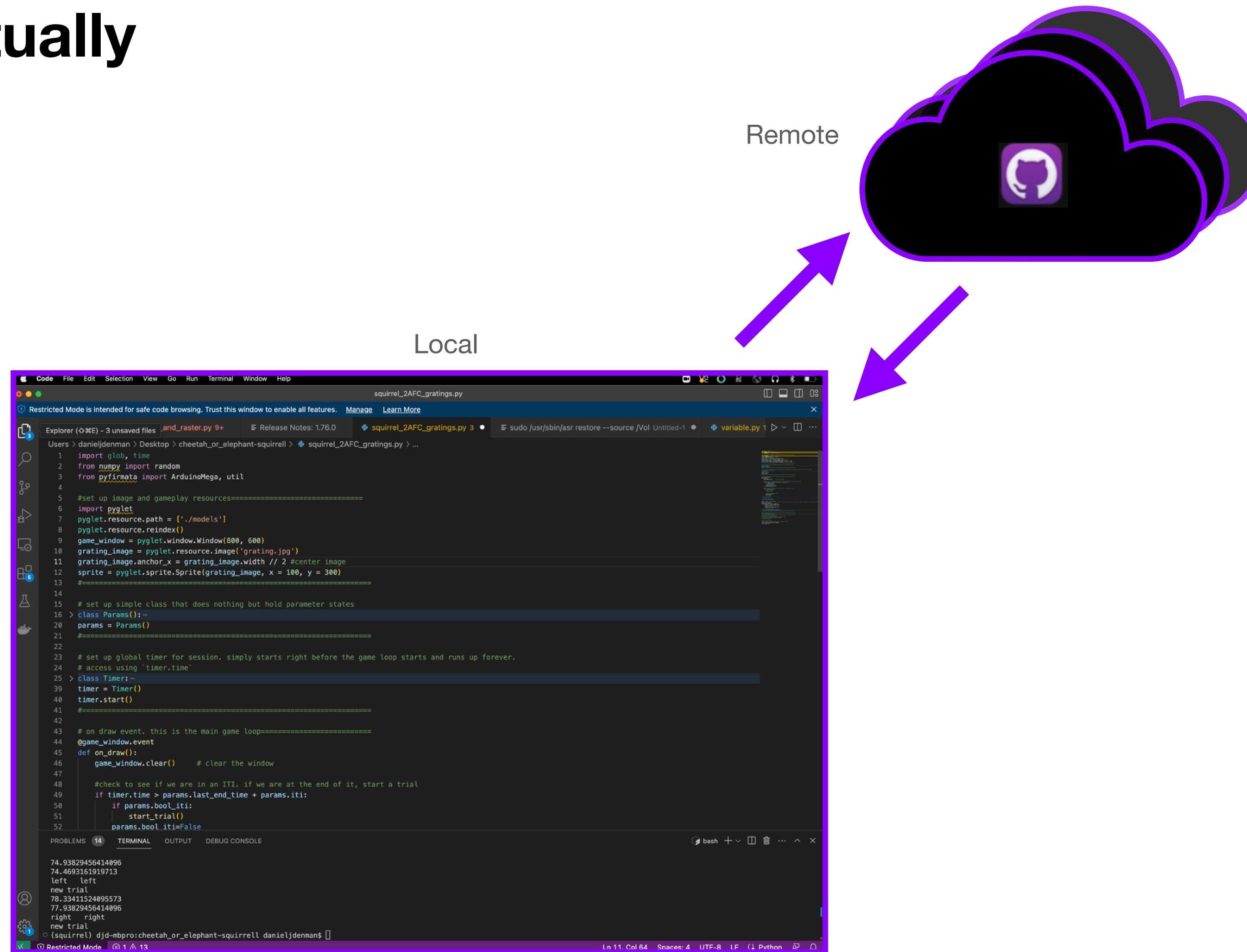
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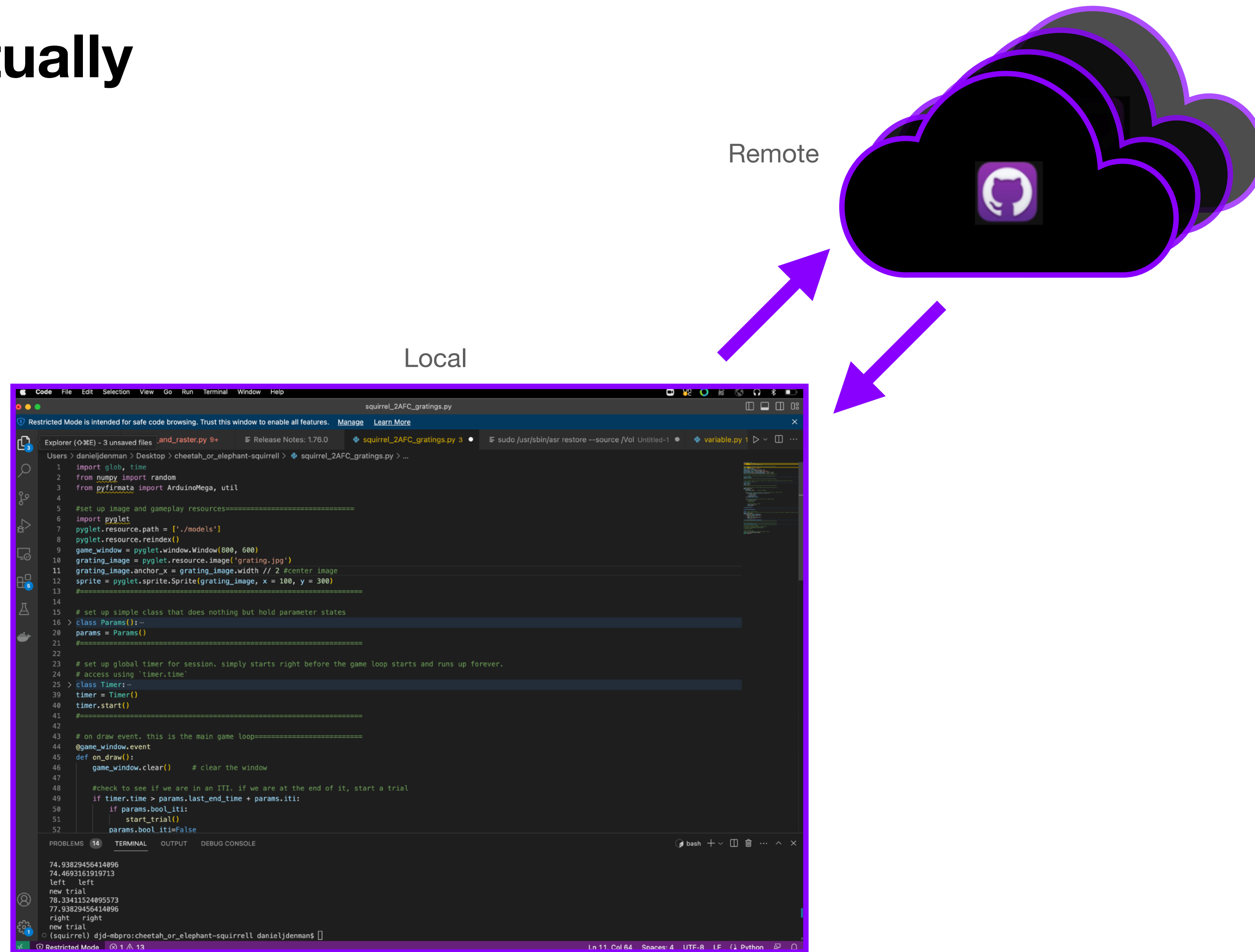
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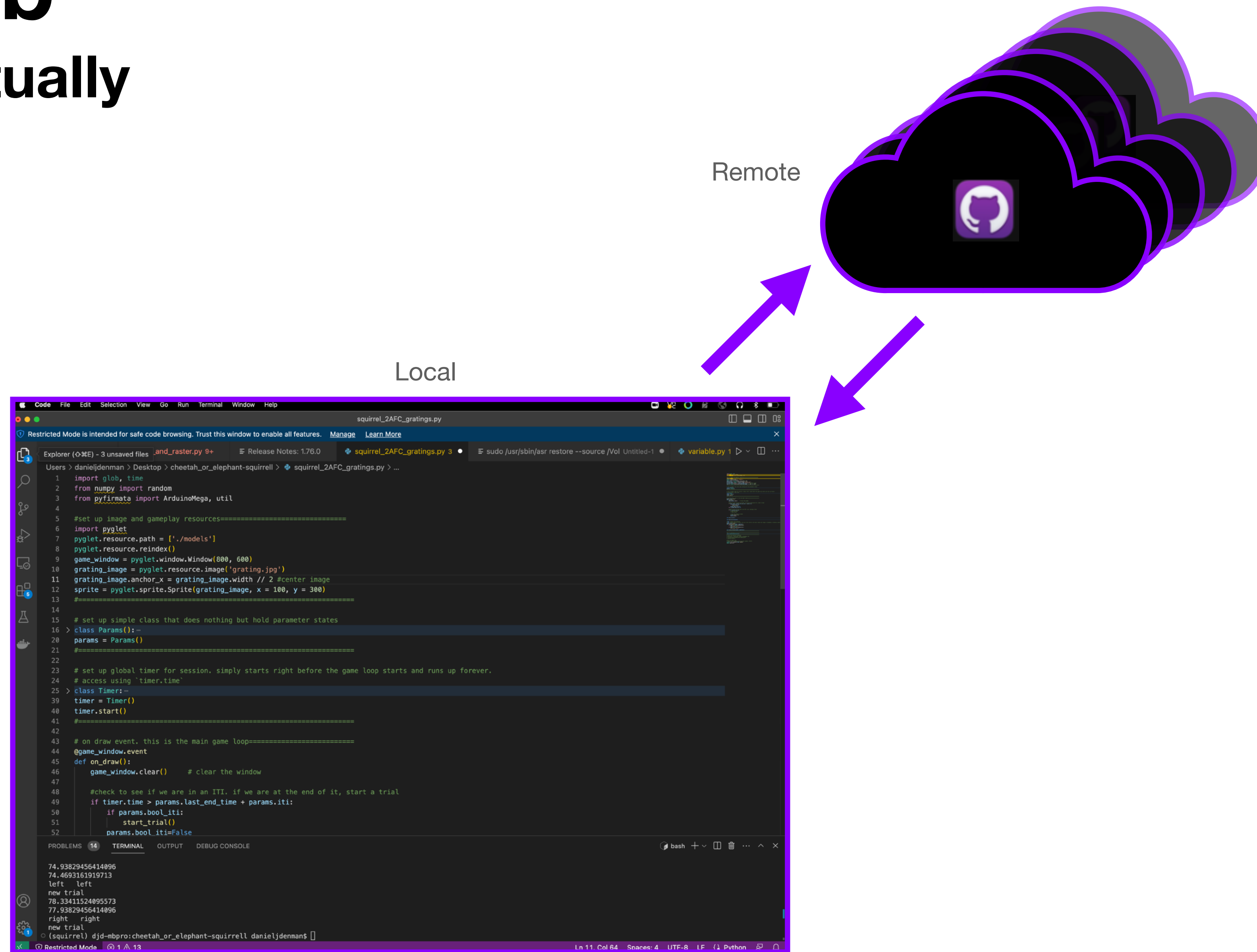
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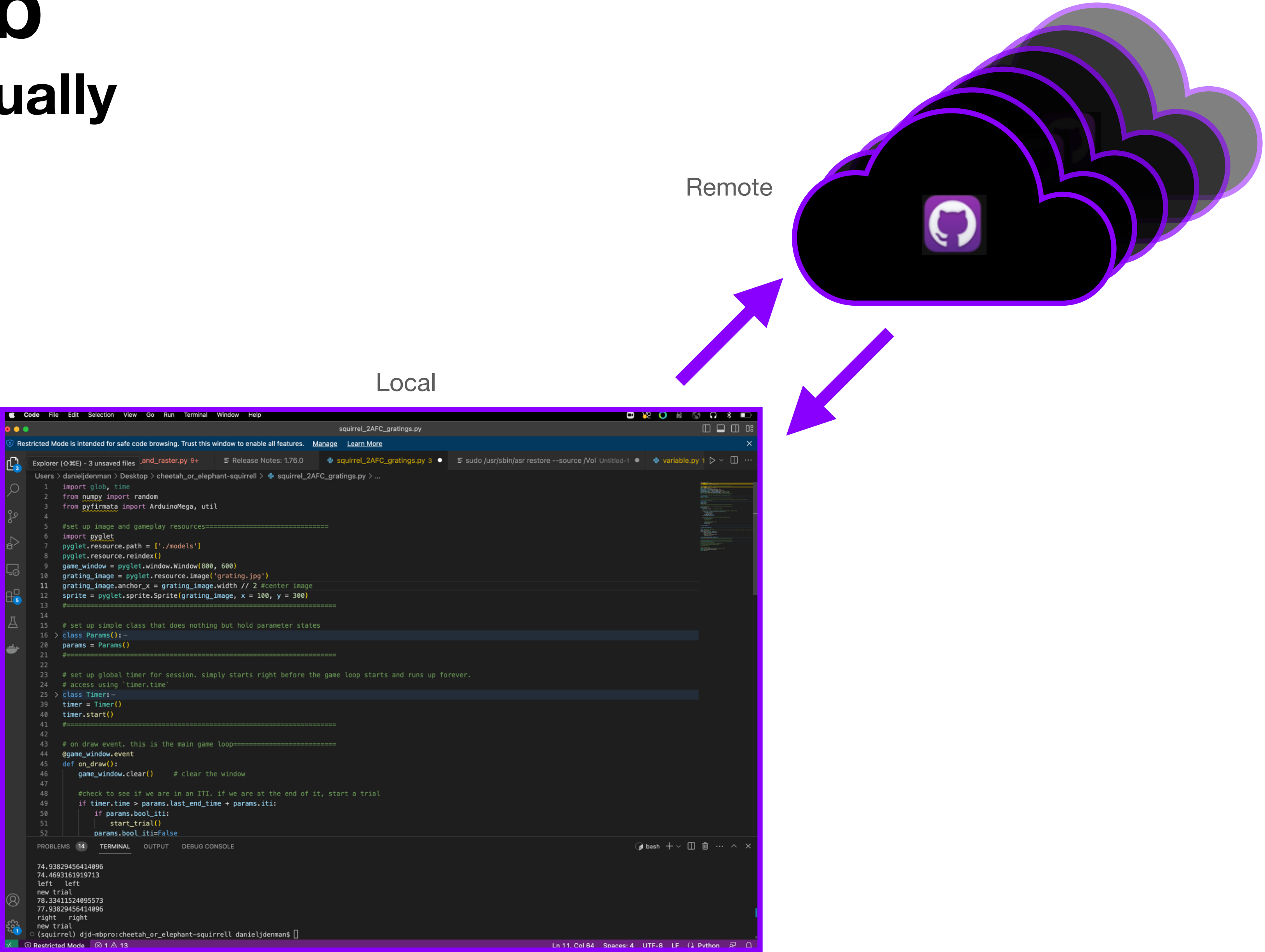
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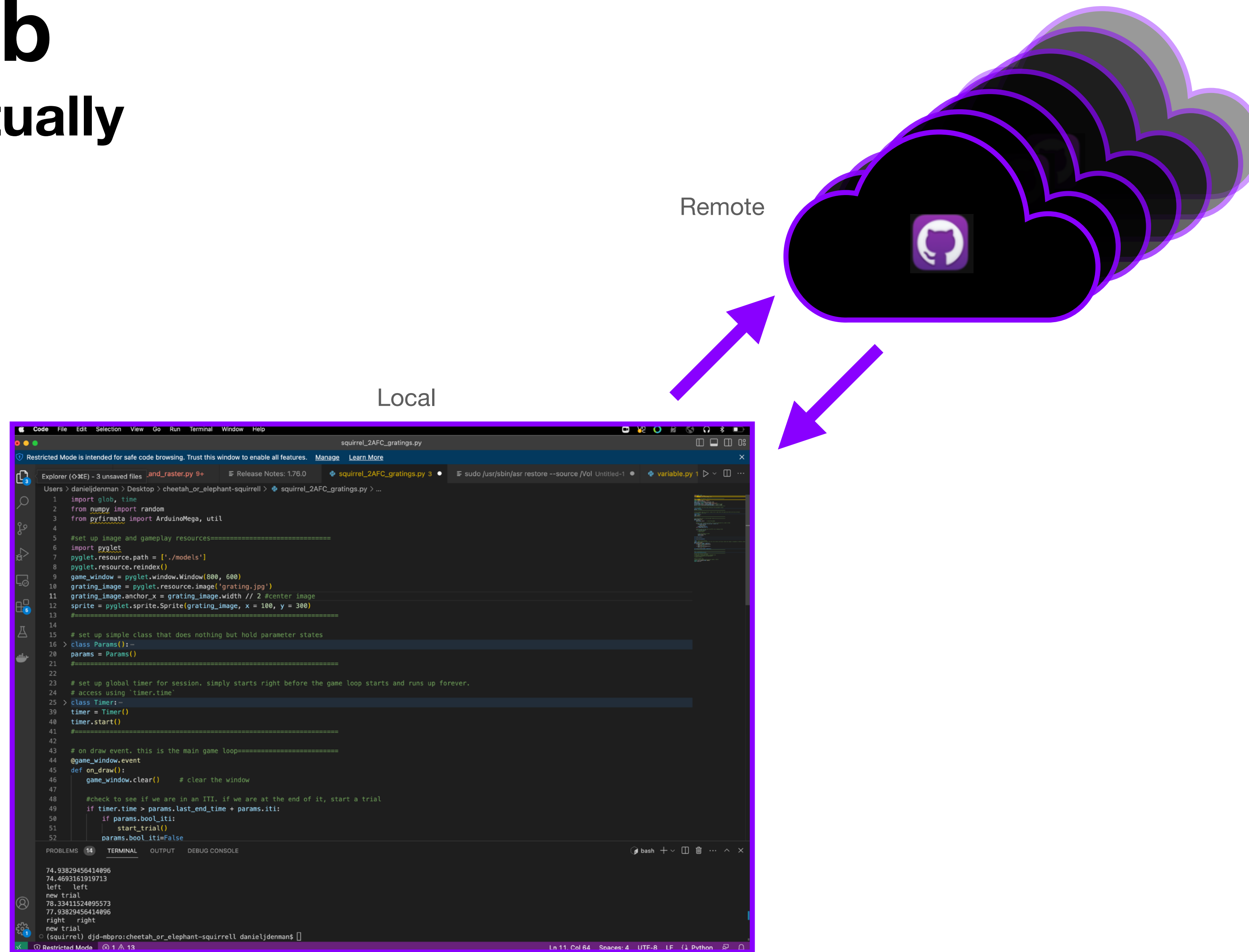
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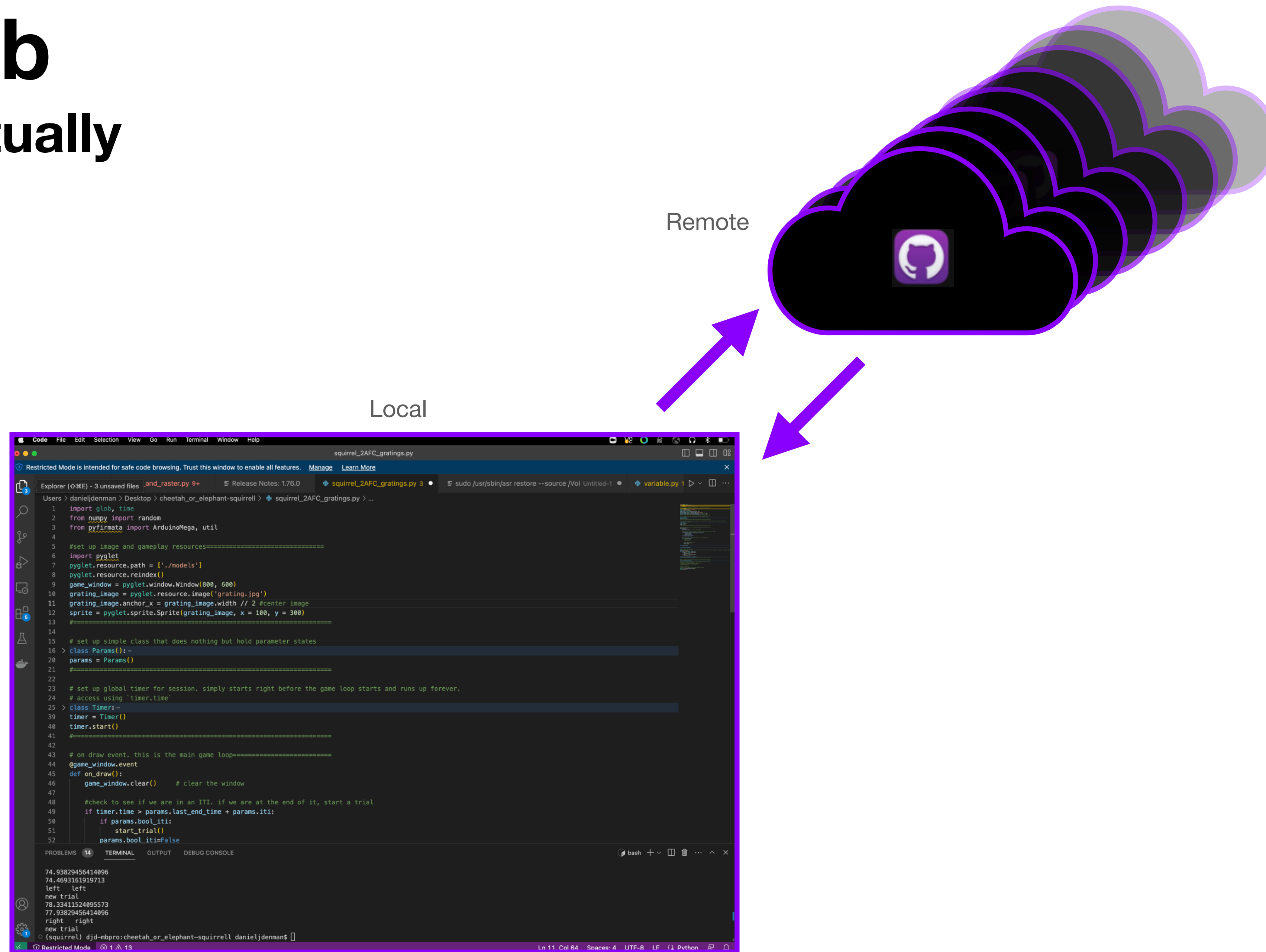
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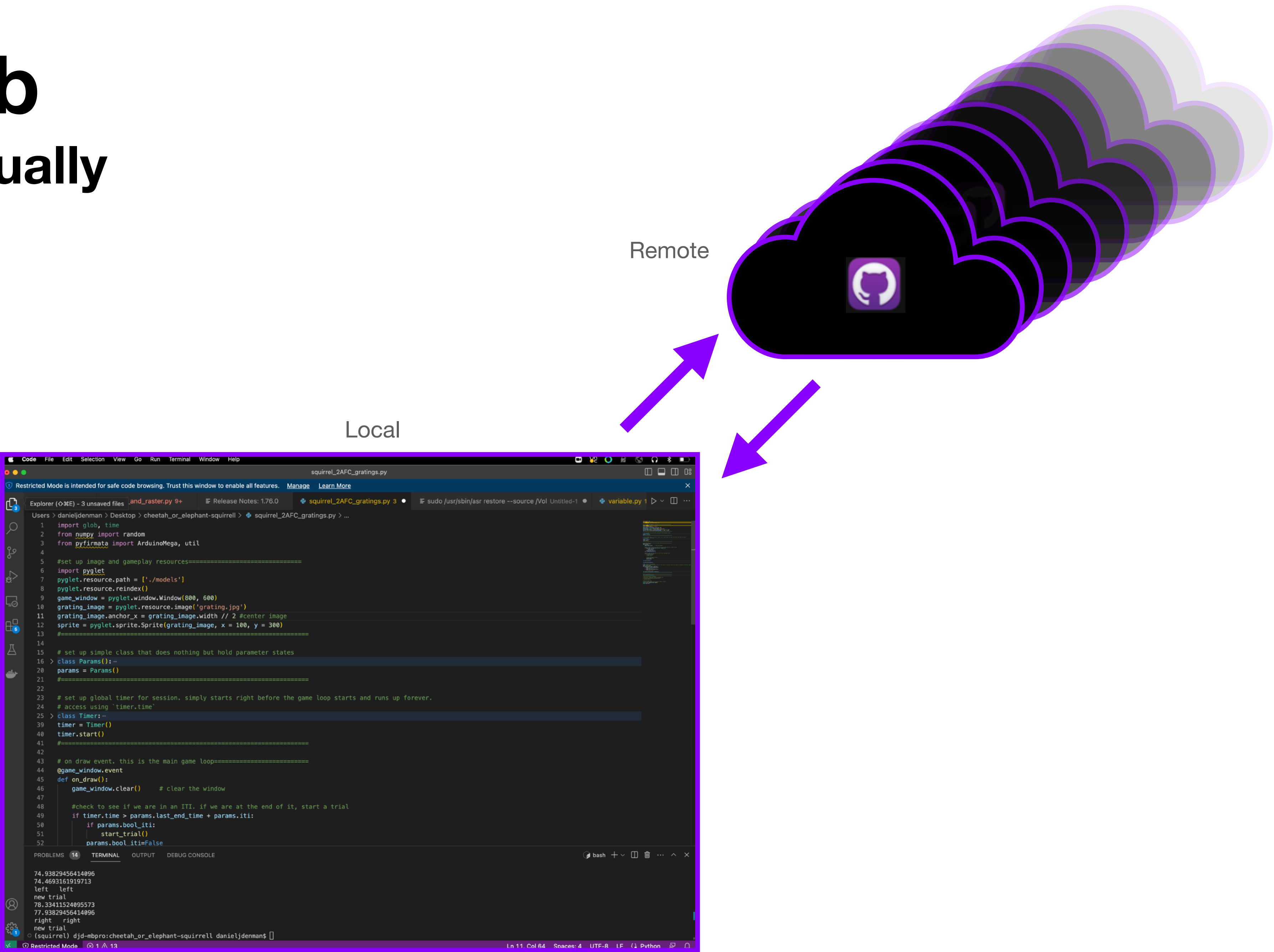
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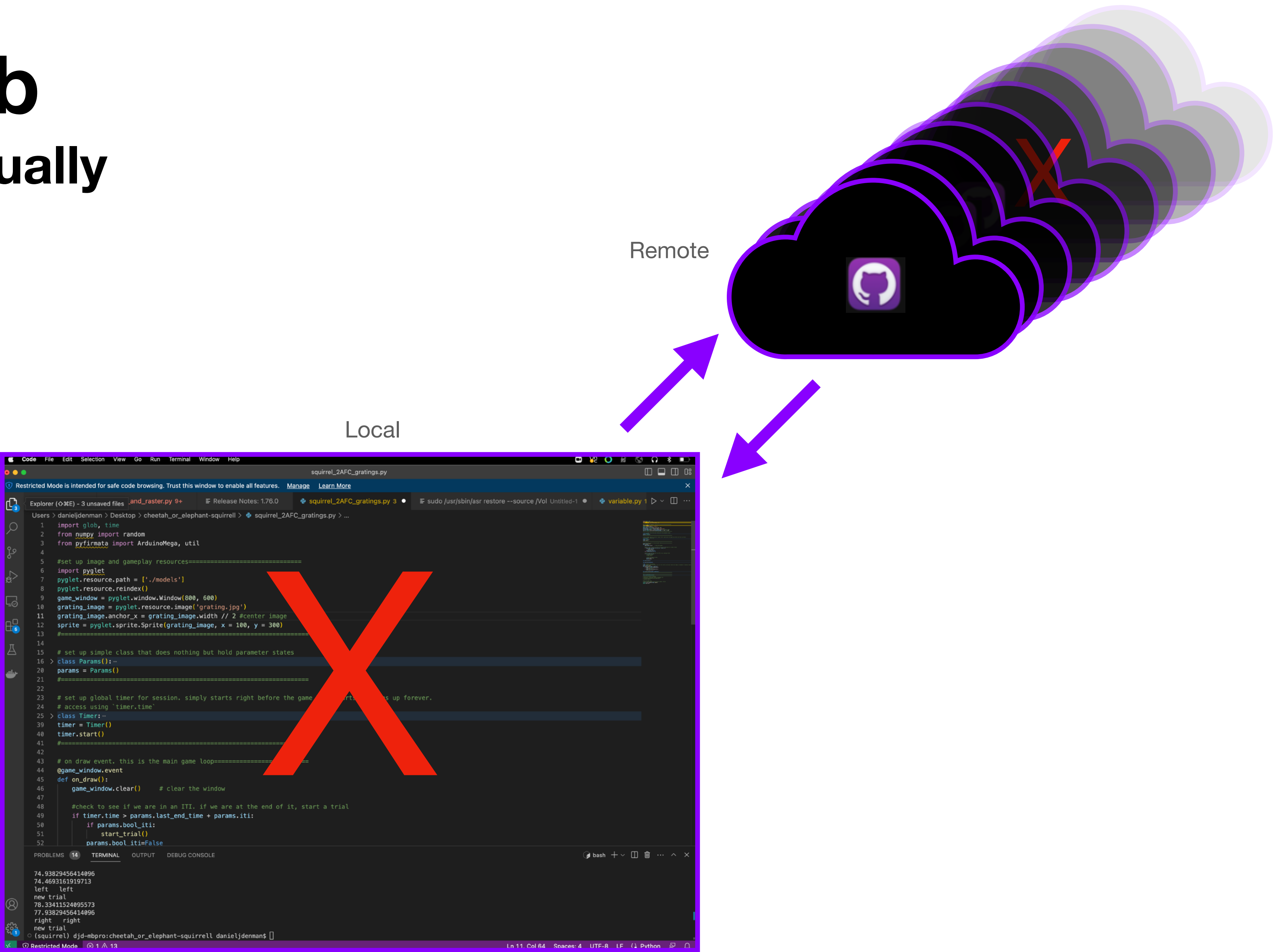
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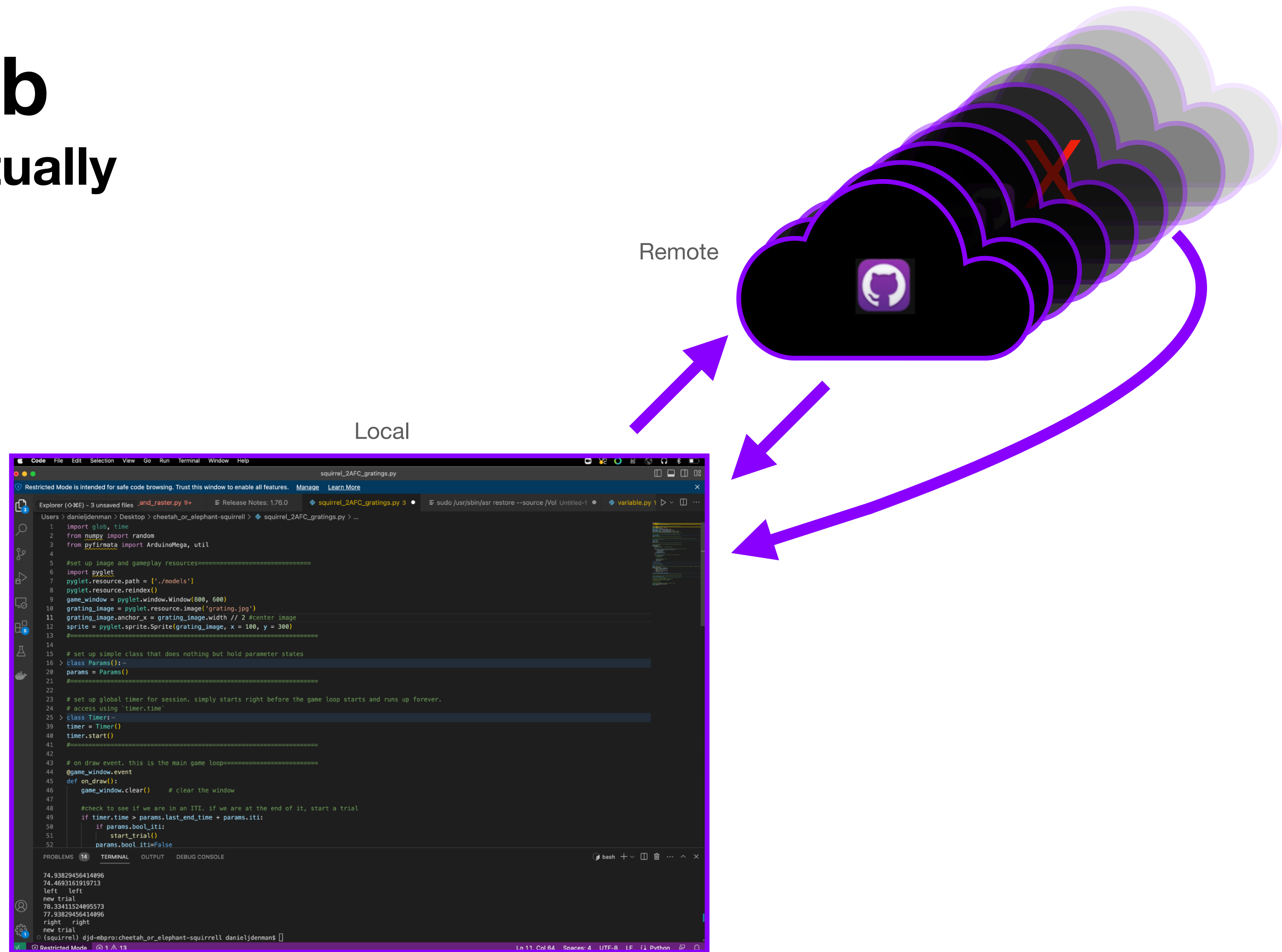
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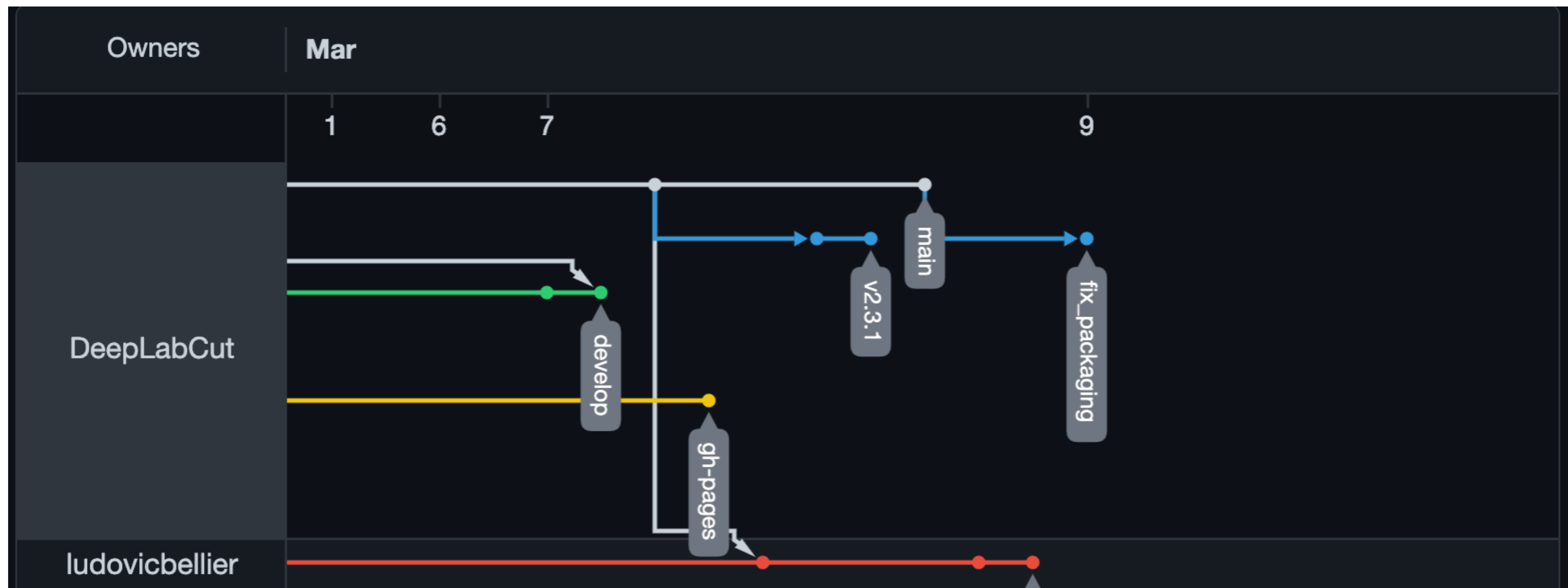
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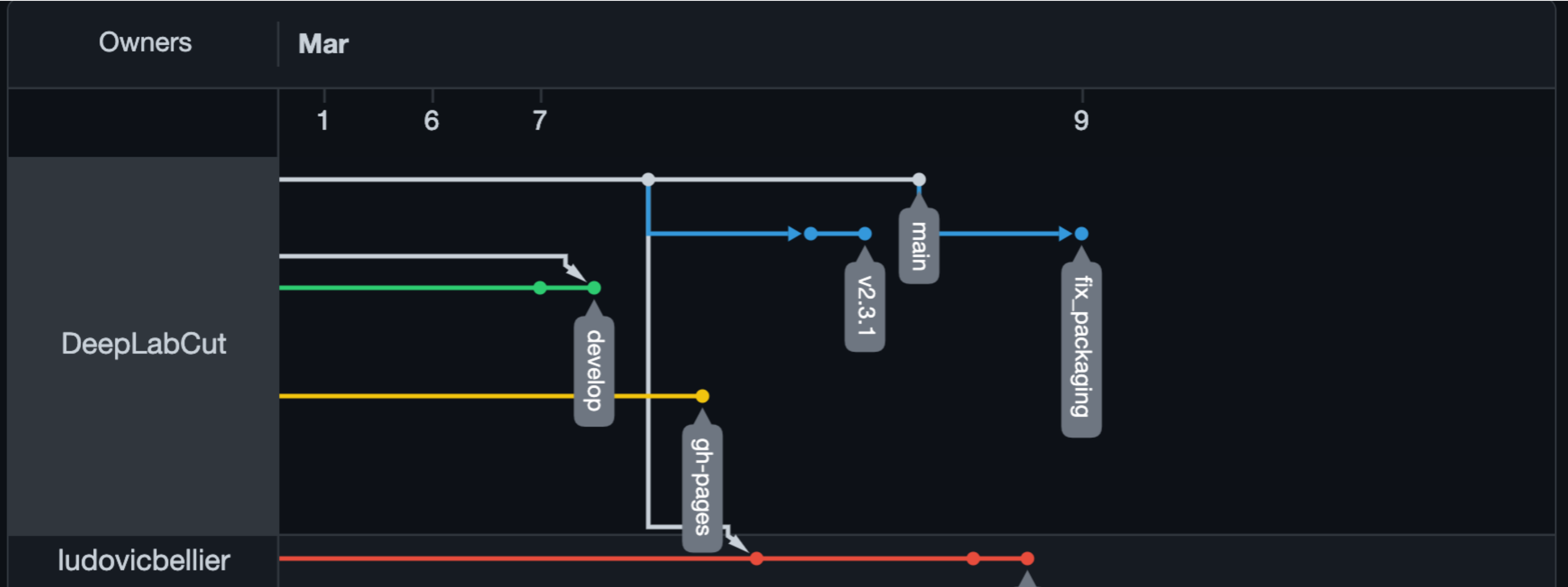
# Github Desktop

## Branching / Forking



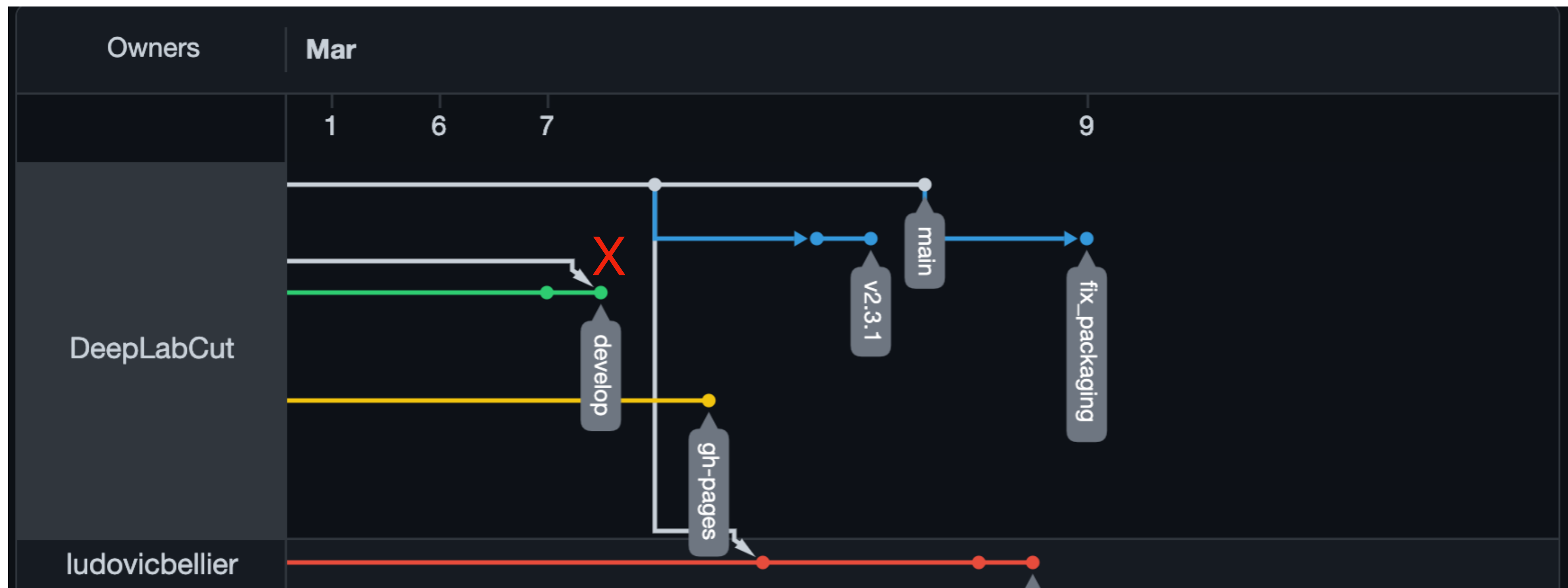
# Github Desktop

## Merging



# Github Desktop

## Conflicts



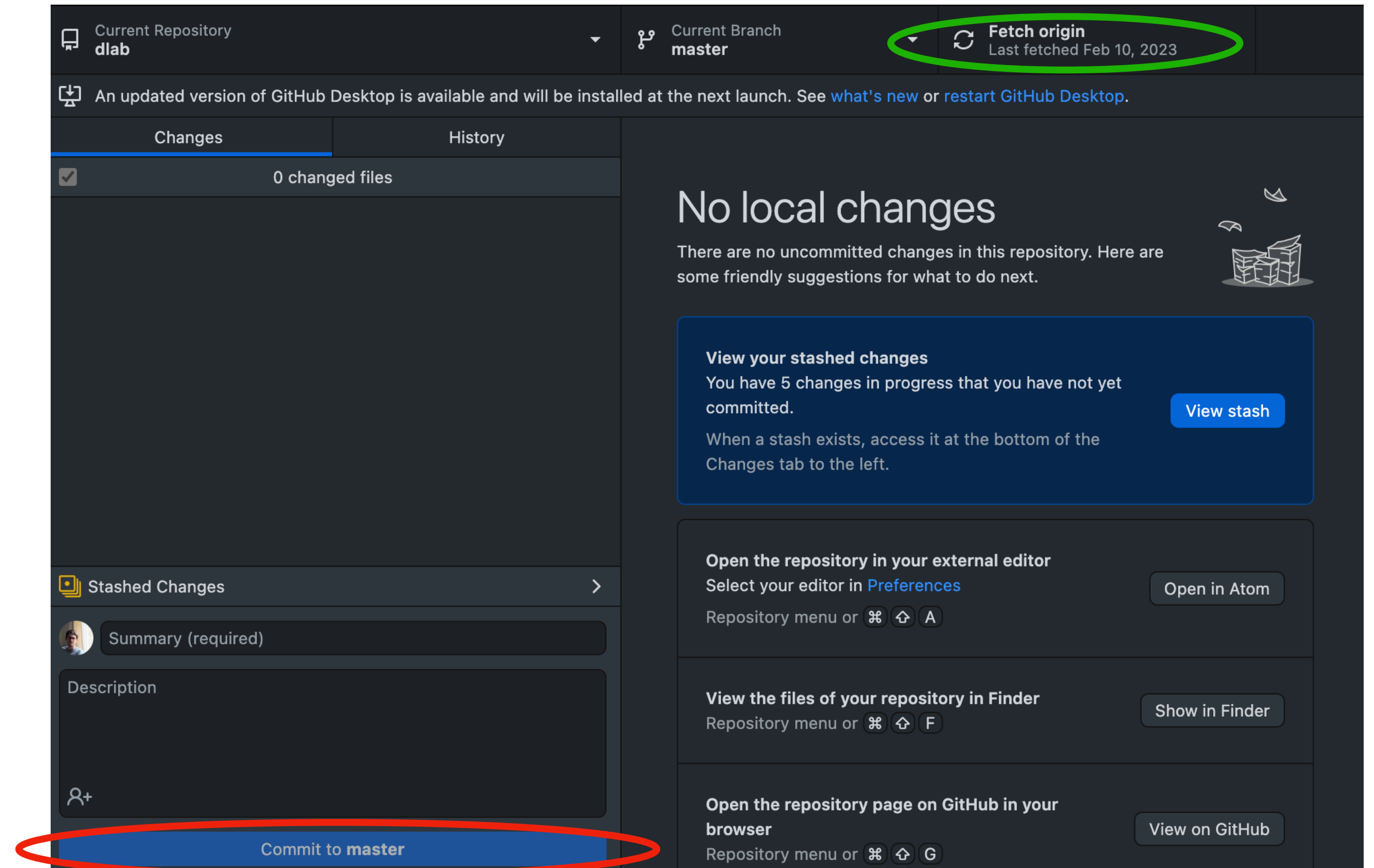


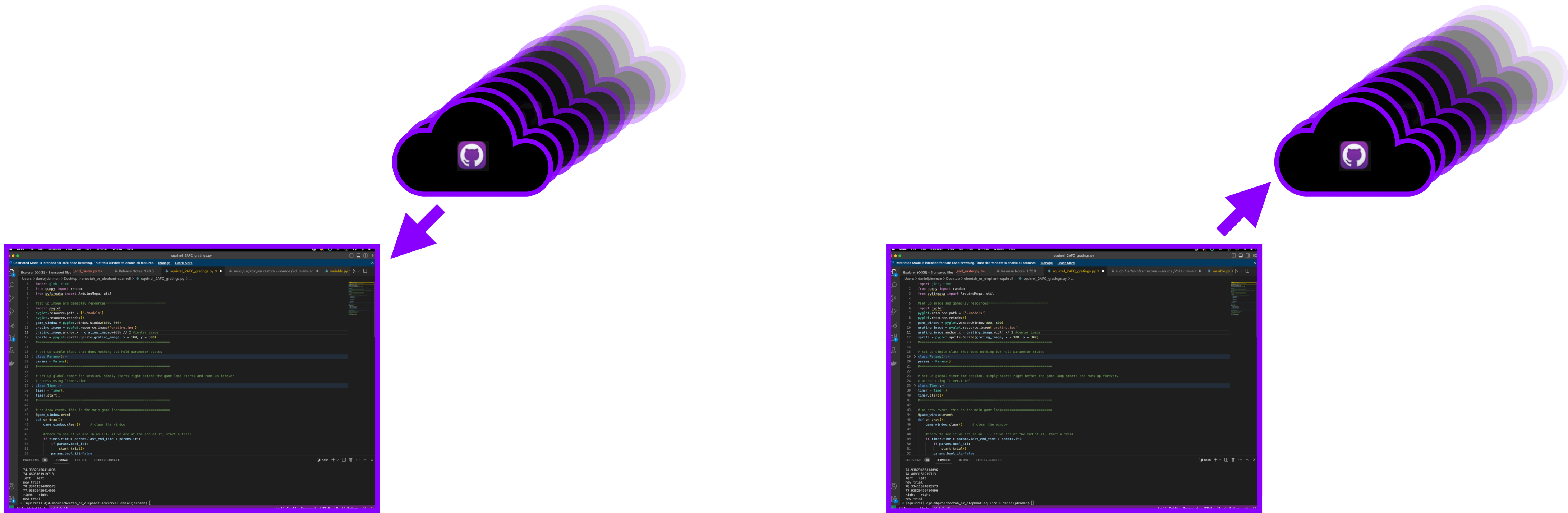
# GitHub

## CLI vs. GUI

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```





```
git fetch origin  
git reset --hard origin/master
```



Break; if you don't have anaconda, install it



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